S2.4 From models to materials

Properties of a material

What you should know

By the end of this section you should be able to:

- Describe bonding as a continuum between the ionic, covalent and metallic models.
- Use bonding models to explain the properties of a material.

In this topic so far, we have looked at ionic, covalent and metallic models as a way to categorise the arrangement of atoms based on electrostatic interactions. Throughout the twentieth century, many chemists used the models of bonding to categorise substances and link the three bonding models together. **Interactive 1** shows some of the examples that were developed. Note that many of the models, terms and symbols used no longer align with those used today. As you scroll and look through the historical methods of categorisation, identify similarities and differences with what you have learned about ionic, covalent and metallic models of bonding.

Interactive 1. Bonding Models and Categorisation of Substances Based on Structure.

More information for interactive 1

In this section, we will discuss the utility and limitations of these three bonding models in explaining the properties of a material. As we think about how to explain the properties of a material consider the following:

- How are the three bonding models useful?
- Are there any limitations to these three bonding models?
- Do you think all substances can be neatly classified into one of the three bonding models?

Chemical bonding

So far, we have classified chemical bonding based on the electrostatic interactions between charged particles:

- Ionic – between oppositely charged ions.
- Covalent – between a shared pair of electrons and positively charged nuclei.
- Metallic – between cations in a lattice and delocalised electrons.

What type of elements (metal/non-metal) would you select that combine to form each type of bonding? We can use a bonding triangle to classify the type of chemical bonding that occurs based on the type of elements that make up the substance, as shown in **Figure 1**.

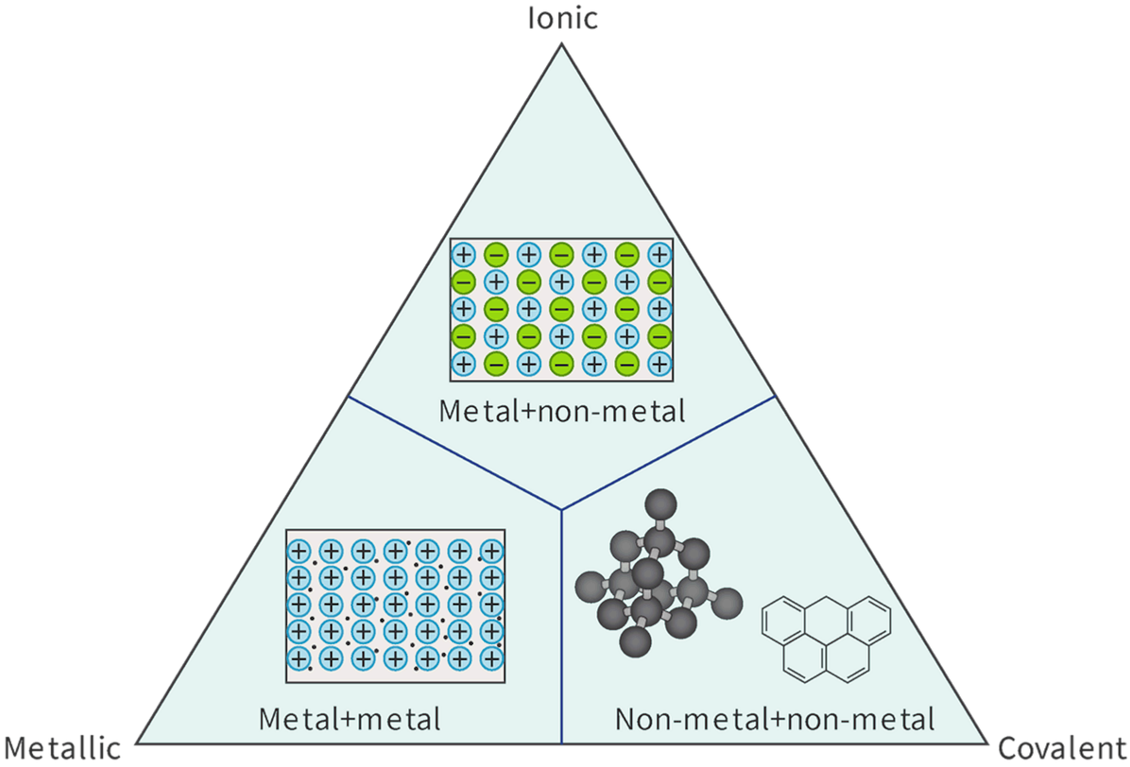


Figure 1. Triangle representing the three types of bonding.

More information for figure 1

The image is a triangle diagram representing three types of chemical bonding based on the elements involved. The triangle is divided into three zones:

1. **Metal + Metal** (Bottom left corner): This section represents metallic bonding, depicted with a grid of positive symbols, indicating metal ions in a sea of electrons.

2. **Metal + Non-Metal** (Top section): This part of the triangle shows ionic bonding, illustrated with a grid-like pattern containing alternating positive and negative symbols, suggesting the attraction between metal cations and non-metal anions.

3. **Non-Metal + Non-Metal** (Bottom right corner): This zone corresponds to covalent bonding. It includes molecular structures, such as a cluster of spheres and a hexagonal shape, symbolizing shared electron pairs between non-metal atoms.

The triangle serves to categorize chemical bonds by placing them at their respective vertices, based on the elements' metallic or non-metallic nature.

[Generated by AI]

This diagram allows classification of a bond between two elements by placing it on a vertex on the triangle, depending on whether the elements are metals or non-metals. If the elements making up a chemical bond are both metals, the bonding is classified as metallic. If the chemical bond is made up of a metal and non-metal, the bonding is classified as ionic. Finally, if the bond is made up of two non-metals, the bonding is classified as covalent. A more complex version of this triangle can be found in section 17 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/triangular-bonding-diagram-van-arkelketelaar-id-45119/) of the DP chemistry data booklet. We will look more extensively at the features of this triangle in section S2.4.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bonding-triangle-id-43843/).

Notice that the three bonding models are placed together in one triangle instead of separating them entirely. What would be the purpose of putting them together? What do you think happens along the boundaries between two bond types? Is it possible for a bond to have characteristics that fit in more than one category?

International Mindedness

The boundary lines that divide ionic, covalent and metallic bonding can be viewed in parallel to the border boundaries that divide nations, territories, districts, states and provinces. Border communities often exhibit similarities in culture, language, food, etc. on both sides of the boundary.

Bonding as a continuum

In the triangle model in **Figure 1**, the edges of the triangle would represent bond types of substances that fall somewhere between two bond types. Can you think of an example of a bond type that falls somewhere in between covalent and ionic bonding? **Figure 2** shows an example of a bonding continuum that exists along the edge of the triangle between covalent and ionic bonding. This bonding continuum shows the transition between pure covalent bonding (e.g. Cl_2) to ionic bonding (e.g. NaCl), with polar covalent bonding lying in between (e.g. HCl).

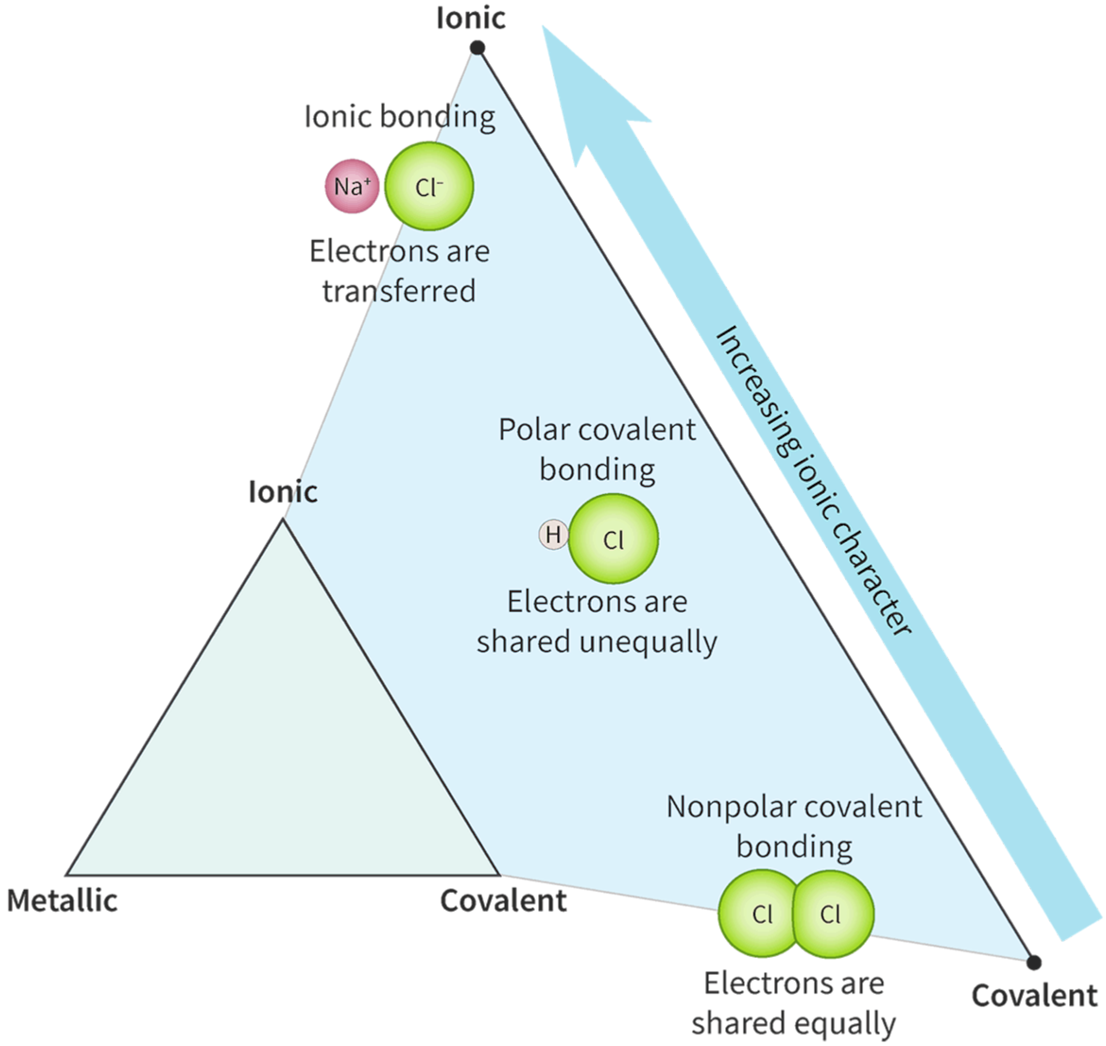


Figure 2. Bonding continuum between covalent and ionic bonding.

More information for figure 2

This diagram illustrates the bonding continuum between covalent and ionic bonding within a triangular model. The left side features Na+ and Cl- ions labeled as ionic bonding, with a note stating that electrons are transferred. The right side has a gradient labeled "Increasing ionic character," showing polar covalent bonding between H and Cl, where electrons are shared unequally. At the bottom, nonpolar covalent bonding is depicted with two Cl atoms. This triangular representation links ionic, polar covalent, and nonpolar covalent bonds along the continuum.

[Generated by AI]

A bonding continuum similarly exists from metallic to ionic bonding, and from metallic to covalent bonding. This means that bonding is best described as a continuum between the ionic, covalent and metallic models rather than discrete categories.

Nature of Science

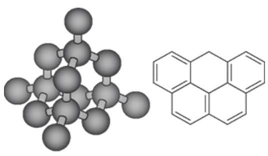
Aspect: Models

It is important to remember that while models can be useful all models have limitations. In the triangle model used to represent bonding in this section, it is a useful model because it allows bonding to be viewed as a continuum rather than discrete. However, the limitation of this particular model is that it does not help you identify which bonds would lie on the extremes (vertices of the triangle) and which bonds would lie between two types of bonding (centre of an edge on triangle). Highlighting the limitations of a model allows for new models to be developed that can more accurately and completely represent natural phenomena.

Properties of a material using bonding models

The physical characteristics of materials used in our everyday lives are closely related to the structure and bonding of the elements. Try out **Interactive 2** below to see whether you can recall the bonding and structure of each of the three types of chemical bonding. Drag the descriptions of the bonding and structure to match the images in the table.

	Metallic	Covalent	Ionic
Bonding description			
Description of structure			
Structural diagram			

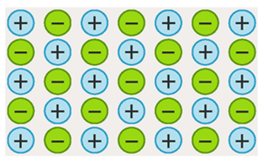


Lattice structure

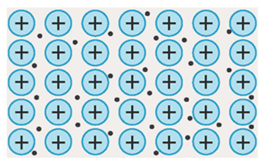
Electrostatic attraction between shared pairs of electrons and positively charged nuclei

Small or large molecules, covalent network structures

Electrostatic attraction between oppositely charged ions



Electrostatic attraction between positive metal ions and delocalised valence electrons



Lattice structure

Check

Interactive 2. Bonding and Structure Descriptions and Images.

More information for interactive 2

An interactive table with missing fields where users need to drag and drop the bonding descriptions, descriptions of structure, and structural diagrams of the metallic, covalent, and ionic bonds.
The drag-and-drop options are as follows:
A structure on the left shows a 3D model representing a diamond structure with a tetragonal arrangement. Each carbon atom is bonded to four other carbon atoms. To the right, a chemical structure features five fused benzene rings, three on the top and two on the bottom.
Lattice structure
Electrostatic attraction between shared pairs of electrons and positively charged nuclei
Small or large molecules, covalent network structures
Electrostatic attraction between oppositely charged ions
A structure illustrates an alternating arrangement of positive and negative ions
Electrostatic attraction between positive metal ions and delocalised valence electrons
A structure of positively charged metal ions surrounded by a sea of electrons
Lattice structure
This interactivity helps users to identify and differentiate between metallic, covalent, and ionic bonds.
Read below for the solution:
Bonding description:
Metallic: Electrostatic attraction between positive metal ions and delocalised valence electrons.
Covalent: Electrostatic attraction between shared pairs of electrons and positively charged nuclei.
Ionic: Electrostatic attraction between oppositely charged ions.
Description of structure:
Metallic: Lattice structure.
Covalent: Small or large molecules, covalent network structures.
Ionic: Lattice structure.
Structure diagram:
Metallic: A structure of positively charged metal ions surrounded by a sea of electrons.
Covalent: A structure on the left shows a 3D model representing a diamond structure with a tetragonal arrangement. Each carbon atom is bonded to four other carbon atoms. To the right, a chemical structure features five fused benzene rings, three on the top and two on the bottom.
Ionic: A structure illustrates an alternating arrangement of positive and negative ions.

The three bonding models can be used to explain the properties of a material. What are some physical properties that we have looked at so far? Which ones are unique to ionic bonding? Covalent bonding? Metallic bonding? Which type of bonding is the most difficult to assign specific physical properties to?

Covalent bonding is usually the most difficult to assign specific physical properties because these properties depend on intermolecular interactions and whether or not the substance is a small molecule, large molecule or covalent network solid.

Materials chemists tend to classify materials according to the structure and bonding of the elements making up a substance; however, many scientists and engineers choose to classify materials based on their physical characteristics. Many of these physical characteristics are the physical properties we looked at in section S2.1.3b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-of-ionic-compounds-id-43559/) for ionic compounds, section S2.2.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) for covalent network structures and section S2.3.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/) for metals. **Figure 3** is a summary of some of the common physical properties used to classify materials.

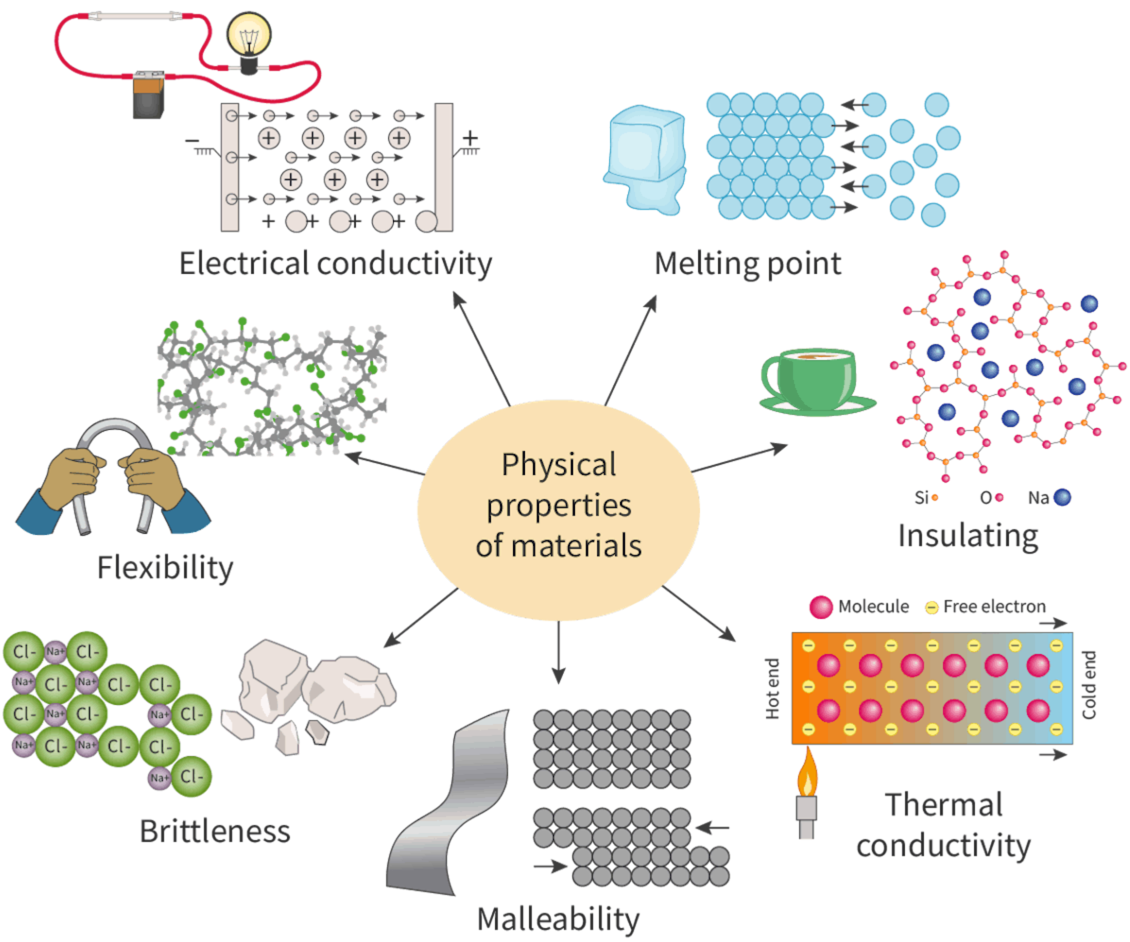


Figure 3. Physical properties of some materials.

More information for figure 3

The diagram illustrates various physical properties of materials with examples. At the center is a beige circle labeled "Physical properties of materials," with lines extending to depict different properties.

1. Electrical Conductivity: Illustrated with an image of an electric circuit showing a battery connected to a light bulb via conductors, demonstrating current flow.

2. Melting Point: Depicted by ice and a grid of particles transitioning from solid to liquid state.

3. Flexibility: Shown using hands bending a bar, with a molecular structure illustrating flexible bonds.

4. Brittleness: Represented by a material breaking along its crystalline lines, with Na+ and Cl- ions depicted.

5. Malleability: Illustrated by bending a sheet of material, depicted by particles rearranging under pressure.

6. Insulating: Shown with a ceramic cup and a molecular structure where Si, O, and Na are connected, highlighting resistance to heat or electricity flow.

7. Thermal Conductivity: Depicted by a bar with a hot and cold end, showing molecules and free electrons transferring heat between ends.

[Generated by AI]

Do you notice any physical properties listed that were not discussed when we looked at ionic, covalent and metallic bonding? Flexibility is the ability of a material to be bent without breaking. This property is similar to malleability. Flexible materials can return to their original shape after experiencing a compression whereas malleable materials remain bent. Insulating is when a material does not permit thermal energy or an electric current to flow freely. This is usually due to the absence of freely moving particles.

When designing a material you would need to ask yourself what properties you would want from the material. The questions below are examples of those that might be asked by someone designing a material object. For each of the questions, consider what type of bonding would have that property.

- Does it need a high melting point?
- Does it need to be electrically conductive in the solid state?
- Does it need to be hard?
- Will it shatter if you drop it?

Kog

- Is it shiny?
- Does it bend easily?
- Do you need it to insulate or conduct heat (thermal energy)?

We know that the physical properties of a material are dependent on the chemical structure and bonding that make it up. In this activity you will use your experiences with everyday objects to predict the bond type you would expect, followed by reviewing explanations for the common physical properties of ionic, metallic and covalent substances.

Activity

- **IB learner profile attribute:** Communicators
- **Approaches to learning**: Social skills — Clearly communicating complex ideas in response to open-ended questions
- **Time required to complete activity**: 30—40 minutes
- **Activity type:** individual activity to start then pair activity to follow

Copy and complete the following table for the objects listed. List all the physical properties that you think the object would have (2—5 properties per object). Looking at your list of physical properties, assign the type of bonding you would expect the elements making up this object to have. For each bonding type assigned, provide a percentage to show how confident you are in your estimated bonding type (e.g. if you were deciding between two bonding types you might list 50% confident, if you were really certain about the bonding type you might list 100% confident).

Object	Physical properties of object	Estimated bonding type	P
Silver spoon 			

Object	Physical properties of object	Estimated bonding type	P
Eggshell 			
Latex balloon 			
Brick 			

Object	Physical properties of object	Estimated bonding type	P
Pencil lead 			
Stainless steel frying pan 			

With a partner, compare your tables.

Did you and your partner have any similarities in a bonding type you were both **confident** in assigning? If so, which bonding type? Discuss reasons why you found it **easy** to assign this bonding type.

Did you and your partner have any similarities in a bonding type you both **lack confidence** in assigning? If so, which bonding type? Discuss reasons why you found it **difficult** to assign this bonding type.

In this next part of the activity you will explain, orally, to your partner why ionic compounds or metallic substances have the physical properties that they do. One partner will explain the physical properties for ionic compounds while the other partner will explain metallic substances.

Ionic compounds are...	Metallic s
Brittle	M
Insulating	C

Kog

Ionic compounds are...	Metallic s
Solids at room temperature	Solids at r

Covalent materials are more difficult to assign physical properties to because the properties can vary depending on how atoms are arranged, whether there are intermolecular forces or covalent bonds holding the solid together, and whether there are free electrons within the structure.

With your partner, brainstorm an example of a covalent solid for each row that would have the physical property stated:

Physical property	Example o
Electrically conductive	
High melting point	
Hard	
Soft	

S2.4 From models to materials

Bonding triangle

Learning outcomes

By the end of this section you should be able to:

- Determine the position of a compound based on its position in the bonding triangle from electronegativity data.
- Predict properties of a compound based on its position in the bonding triangle.

Chemists have long desired the ability to classify substances based on their bond types to explain physical properties. As we saw in section S2.4.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/) chemical bonding does not exist in discrete categories but rather as a continuum between ionic, covalent and metallic models. Anton van Arkel and Jan Ketelaar were two chemists that attempted to classify bonds along a bonding continuum by adding more detail to the simple bonding triangle that we looked at earlier. **Interactive 1** below shows two examples of these bonding triangles. Scroll through the photos to view them.

Interactive 1. Van Arkel's 1941 Bonding Triangle

More information for interactive 1

The slide 1 in the interactive is Van Arkel's bonding triangle, a model developed to classify different types of chemical bonds based on the properties of elements and compounds. This triangle is divided into three main bonding types: iconic bond (top of the triangle), covalent bond (bottom left corner) and metallic bond (bottom right corner).
Ionic Bonds — Formed when electrons are completely transferred from one atom to another, creating charged ions. Example mentioned in the triangle: CsF (Caesium Fluoride).

Covalent Bond — Formed when atoms share electrons to achieve stability. Examples mentioned in the triangle: F_2 (fluorine gas) and I_2 (iodine gas).
Metallic Bonds — Found in metals, where electrons move freely among positively charged ions. Example mentioned in the triangle: Sn (Tin).
Compounds along the edges show a mix of bonding characteristics. BF_3, CF_4, and NF_3 lie between ionic and covalent bonds. AlP, ZnTe, and SnPb are mentioned between metallic and covalent bonds. CoS is between ionic and metallic bonds, showing both properties.
This diagram helps students understand bonding variations and bond types, rather than treating them as separate categories.
Slide 2 interactive displays Ketelaar's bonding triangle, a model that classifies chemical bonds based on the properties of different compounds. The triangle is divided into three main bonding categories:
Metallic Bonds (Top Corner) — Found in metals, where electrons move freely. Example mentioned in the triangle: Li (Lithium), Ag (Silver), Sn (Tin).
Covalent Bonds (Bottom Left Corner) — Occur when atoms share electrons to form stable molecules. Examples mentioned in the triangle: F_2 (Fluorine), I_2 (Iodine), ClF (Chlorine monofluoride).
Ionic Bonds (Bottom Right Corner) — Formed when electrons are transferred between atoms, creating charged ions. Example: CsF (Caesium).
This diagram allows users to see that bonding falls on a spectrum that includes covalent, ionic, and metallic bonds.

The examples of bonding triangles shown in **Interactive 1** is a way to classify specific bonds. While such diagrams are useful, they are an estimate of the relative location of a substance on the bonding continuum, based on observed properties. Can you think of any ways to quantify a diagram of this type? Do you know of any numerical parameters that describe bond characteristics?

When we considered the bonding continuum earlier in section S2.4.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/properties-of-a-material-id-43465/) we looked at how polar covalent bonds are a bond type between ionic and covalent. What numerical parameter can you use to deduce the polar nature of a covalent bond? Recall from section S2.2.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-polarity-id-43853/) that electronegativity is the tendency of an atom to attract a shared pair of electrons. Each atom has an assigned electronegativity value, relative to fluorine, that can be found in section 9 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/) of the DP chemistry data booklet. These values provide an indicator about the degree to which an individual atom will attract an additional electron from another atom.

- Metals have a high tendency to lose electrons, low tendency to attract additional electrons, therefore low electronegativity values.
- Non-metals have a high tendency to gain electrons, therefore high electronegativity values.

How might we use electronegativity values to quantify bond types between two elements? Consider the types of elements and the electronegativity values from each element for each bond type. **Table 1** shows a summary of the types of elements and electronegativities for each bond type.

Table 1. Types of elements and electronegativities for the bond type between two eleme

Bond type	Types of elements	Electronegativity	
Ionic	Metal	Low	Large differ in electronegat
	Non-metal	High	
Covalent	Non-metal	High	High averag electronegat
	Non-metal	High	
Metallic	Metal	Low	Low averag electronegat
	Metal	Low	

Nature of Science

Aspects:

- Models
- Science as a shared endeavour

The original bonding triangles were qualitative until University of Cincinnati professor William B. Jensen began quantifying van Arkel's and Ketelaar's triangles of bonding using electronegativity values for the purposes of teaching his own students. He published a quantified triangle in the 1980s and again in the 1990s, but his name is frequently omitted from the bonding triangle. Different techniques and mathematical analyses are currently being used to further develop bonding triangles by universities around the world. The most common one currently used is the Van Arkel-Ketelaar triangle. You can find a version of this triangle in section 17 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/triangular-bonding-diagram-van-arkelketelaar-id-45119/) of the DP chemistry data booklet and shown below in **Figure 1**.

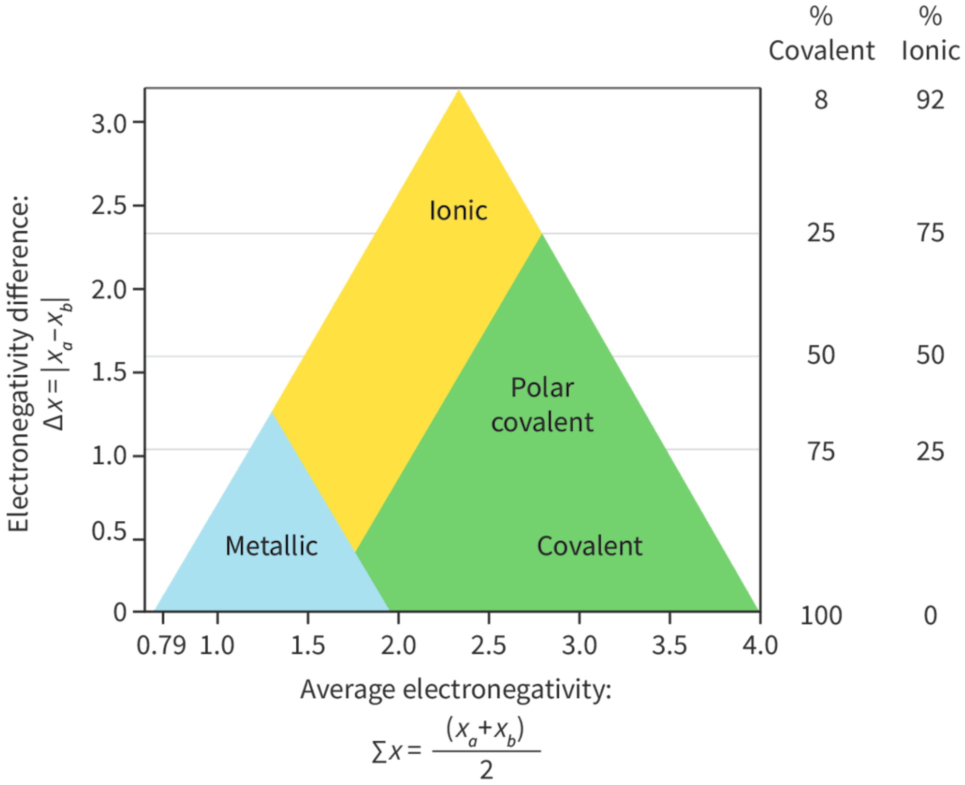


Figure 1. van Arkel-Ketelaar triangle of bonding diagram.

More information for figure 1

The image is a diagram known as the van Arkel-Ketelaar triangle, used to illustrate the types of chemical bonding between two elements based on their electronegativity. The x-axis represents the average electronegativity ((x_a + x_b) / 2) ranging from 0.79 to 4.0, while the y-axis represents the difference in electronegativity (Δx = |x_a - x_b|) ranging from 0 to 3.0.

The diagram is a triangle divided into three regions: "Metallic" at the bottom-left with lower values of both average and difference in electronegativity, "Polar covalent" in the middle, and "Ionic" at the top with higher values. To the right of the triangle, there are percentages indicating the covalent and ionic character ranging from 8% covalent and 92% ionic to 100% covalent and 0% ionic.

[Generated by AI]

This version quantifies the bonding triangle by plotting the average electronegativity between two elements on the *x*-axis and the difference of electronegativity between the two elements on the *y*-axis. Where the point falls within the triangle would be its bond type. While the distinctions between the three bond types are fairly clear on the diagram, the distinction between pure covalent and polar covalent is not. This is where deducing the percentage covalent and ionic character can be useful. You will notice these percentages provided on the right-hand side of the bonding triangle in **Figure 1**.

Let's consider where sodium, hydrogen and sodium chloride would fall on the bonding triangle. We would first need to start by identifying the electronegativities for each element followed by calculating the average electronegativity and the difference in electronegativity. This is shown below in **Table 2**.

Table 2. Electronegativities, difference in electronegativities and averag sodium, hydrogen and sodium chloride.

	Na	H_2
Electronegativity χ_a and χ_b	Sodium 0.9	Hydrogen 2.2
	Bonding occurs between atoms of Na	Bonding occurs between atoms of H
Difference in electronegativity	$\Delta\chi = \lvert\chi_a - \chi_b\rvert$ $\Delta\chi = 0.9 - 0.9$ $= 0$	$\Delta\chi = \lvert\chi_a - \chi_b\rvert$ $\Delta\chi = 2.2 - 2.2$ $= 0$
Average electronegativity	$\sum\chi = \frac{(\chi_a + \chi_b)}{2}$ $\sum\chi = \frac{(0.9 + 0.9)}{2}$ $= 0.9$	$\sum\chi = \frac{(\chi_a + \chi_b)}{2}$ $\sum\chi = \frac{(2.2 + 2.2)}{2}$ $= 2.2$
Bond type see **Figure 2**	Metallic	Covalent

You will notice that for sodium and hydrogen, the bonding occurs between atoms of the same element, resulting in a difference in electronegativity of zero. Both sodium and hydrogen would be plotted on the *x*-axis of the bonding triangle and have a 0% ionic character as shown in **Figure 2**.

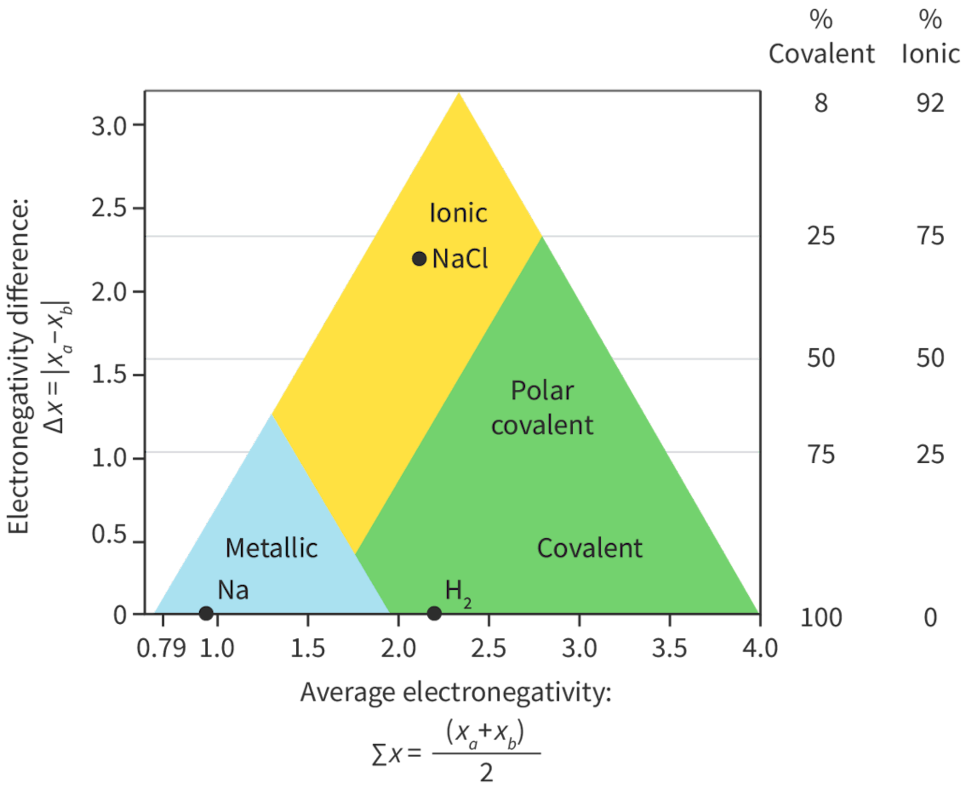


Figure 2. Bonding triangle with Na, H_2 and NaCl plotted onto it.

More information for figure 2

The image is a bonding triangle graph plotting different compounds based on their electronegativity properties. The X-axis represents the average electronegativity labeled as "(x_a + x_b)/2," ranging from 0.79 to 4.0. The Y-axis represents the electronegativity difference, labeled as "|x_a - x_b|," ranging from 0 to 3.0. The triangle is divided into three distinct regions.

1. The bottom left blue area is labeled "Metallic" with "Na" marked there.
2. The top yellow section is labeled "Ionic" with "NaCl" marked there.
3. The right green section is divided into "Polar covalent" and "Covalent" subregions, with "H2" plotted in the "Covalent" area.

Percentages at the right show covalent and ionic character, with 100% covalent at the bottom and 100% ionic at the top.

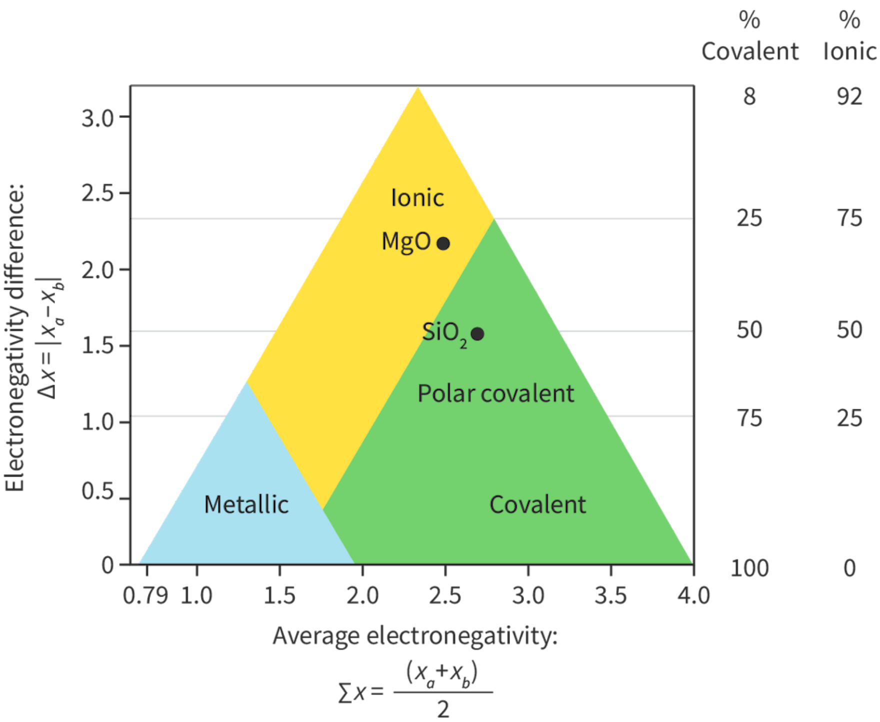
[Generated by AI]

Let’s try another example.

Worked example 1

Silicon dioxide and magnesium oxide are both found within the Earth’s crust. Classify the bond type and estimate the percent covalent character for each.

	SiO_2	
Electronegativity χ_a and χ_b		Oxyger 3.4
	Silicon 1.9	
Difference in electronegativity	$\Delta\chi = 3.4 - 1.9$ $= 1.5$	
Average electronegativity	$\sum\chi = \frac{(3.4 + 1.9)}{2}$ $= 2.65$	
Bond type see diagram below	Covalent	
Percent covalent character	Approximately 50%	



The physical properties of elemental substances, metals and various binary compounds can be explained by considering the structure, bonding and the position in the bonding triangle.

Study skills

It is important to note that we cannot use the bonding triangle for substances made up of more than two elements (e.g. ionic compounds with polyatomic ions and molecules with more than two different atoms). It can only be used for elemental substances and binary compounds.

In the drag and drop interactive, drag the appropriate substances (A–H) into their respective locations in the bonding triangle.

Interactive 2. Where do the Seven Substances go in the Bonding Triangle?

More information for interactive 2

This interactive image displays the **van Arkel—Ketelaar triangle**, a tool used to classify the type of bonding between elements based on electronegativity values.
The diagram plots the electronegativity difference: $\Delta\chi = |X_a - X_b|$ on the y-axis and average electronegativity: $\Sigma\chi = \frac{(X_a - X_b)}{2}$ on the x-axis. The x-axis ranges from 0.80 to 4.0, and the y-axis ranges from 0 to 3.2.
The triangle is divided into **three colored regions** representing different bonding types:

- **Yellow (top left): Ionic bonding**
- **Green (lower right): Polar covalent bonding**

- **Blue (lower left): Metallic bonding**

Users must drag and drop the correct molecules and elements into the corresponding positions in the diagram. The drop zones are provided at the following locations:
Top right of the yellow section
The bottom right of the yellow section
The top center of the green section
The bottom right of the green section
The bottom left corner of the green section
The bottom right corner of the blue section
The bottom left of the blue section

The drag-and-drop options provided are:
A- Magnesium hydride
B - Potassium
C - Water
D - Sodium chloride
E - Oxygen
F - Aluminium
G - Silicon

This interactivity helps users understand how electronegativity differences and averages determine the type of chemical bonding.
Read below for the solution:
Top right of the yellow section - D - Sodium chloride
The bottom right of the yellow section - A- Magnesium hydride
The top center of the green section - C - Water
The bottom right of the green section - E - Oxygen
The bottom left corner of the green section - G - Silicon
The bottom right corner of the blue section - F - Aluminium
The bottom left of the blue section - B - Potassium
This activity helps learners understand how **bond type correlates with electronegativity difference and average electronegativity:**

Theory of Knowledge

We use scientific models to simplify theories and to make predictions. You have just learned how to use the van Arkel-Ketelaar bonding triangle to predict the bonding in binary compounds as metallic, ionic or covalent. The concept of a bonding triangle has persisted in chemistry and many variations have been developed and modified over time. It is possible that in time, the van Arkel-Ketelaar triangle will undergo another change with advancements in knowledge and experimental evidence. Can the use of a changing scientific model reliably predict the characteristics of all compounds?

Now that you have identified the bond types of each of the substances (A–G) drag and drop the associated structure and notable characteristic that corresponds to each substance in this interactive task.

Interactive 3. Identify the Bond Type, Structure and Notable Properties of the Seven Substances.

More information for interactive 3

What might you expect about the bonding and properties of substances that fall along a boundary line? Since bonding is best described as a continuum, substances that fall along boundary lines show intermediate bonding and properties. For example, covalent network solids tend to fall near the metallic and ionic boundaries but still within the covalent region of the triangle. These substances have lattice structures and high melting points like metals and ionic compounds, while still forming covalent bonds between atoms.

SiO_2 is an example of a substance we looked at in the worked example that is in the covalent region but falls close to the ionic bonding boundary line. SiO_2 is an example of a network covalent structure we explored in section S2.2.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) that is brittle, has a high melting point (1700 °C), but still has covalent bonds between atoms of silicon and oxygen.

In this activity, you will have the opportunity to think about how all three types of bonding are important for building structures out of reinforced concrete.

Activity

- **IB learner profile attribute**: Communicator
- **Approaches to learning**: Communication skills — Using digital media for communicating information
- **Time required to complete activity**: 45 minutes
- **Activity type**: Pair activity

Reinforced concrete is a composite material that is an example of a heterogeneous mixture. The different components of a composite material each have different properties that complement one another when combined. It is one of the most widely used building materials globally. The most common substances found in reinforced concrete are:

Component Type	Composition
Concrete	Cement (primary made up of C
	Rocks and aggregates to make concre SiO_2, Fe_2O_3)
Reinforcement	Rebar (primarily made

Complete the following with a partner:

- Assign where each of the substances would fall in the triangle of bonding.
- Estimate the percent ionic character for each substance.
- Discuss with your partner what properties you would expect each substance to have based on the position within the bonding triangle.
- Select a building structure of your choosing from your local community built from reinforced concrete. This could be a bridge, building, dam, pier, stadium, etc.

- Create a piece of media (video commercial, magazine advertisement, podcast, poster) explaining how the atomic structure and bonding of each substance gives rise to the desired properties of your building structure.

S2.4 From models to materials

Alloys

Learning outcomes

By the end of this section you should be able to:

- Define the term alloy.
- Compare how the structure and properties of an alloy differs from its base metal.
- Explain the properties of alloys in terms of non-directional bonding.

When you look around your kitchen, your household appliances, city skyscrapers, car bodies or tools, can you identify what metal many of these things are made from? Many metallic objects commonly encountered are made from steel as shown in **Interactive 1**. Scroll through the photographs to see what they are.

Interactive 1. Examples of Commonly Encountered Materials Made of Steel.

More information for interactive 1

Steel is one of the most commonly used materials in the world. Have you heard of steel before? What about stainless steel? If you look on the periodic table, do you see steel listed as an element? What does this suggest that steel is? Do you know what elements or combination of elements make steel? What is it about the metallic structure and bonding of steel that makes this metal so useful?

Recall from section S2.3.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/) that the metallic bond is the electrostatic attraction between cations in a lattice and delocalised valence electrons. When we looked at ionic and covalent bonding, we studied the bonding between atoms of different elements. In section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) you learned that atoms of different elements bonding together in a fixed ratio forms a compound. But when we looked at metals, we only considered the bonding between atoms of the same element. Why do you think this is?

When metals combine with other metals, they are less likely to form compounds, instead they form something called an alloy. Alloys are mixtures of a metal and other metals or non-metals. In general, alloys are made up of a base metal with additional elements added. This is illustrated in **Figure 1**. The addition of another element results in a metal with different properties than the base metal. Steel is an example of an alloy that is a mixture of iron with small amounts of carbon. Note that we can also use the triangle of bonding we looked at in section S2.4.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bonding-triangle-id-43843/) to analyse the type of bonding for alloys made up of only two elements.

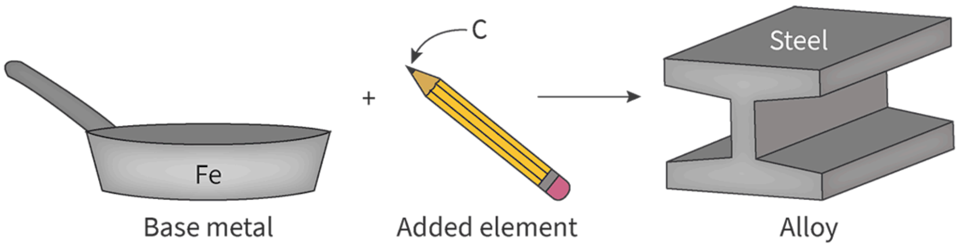


Figure 1. An alloy is made up of a base metal with small amounts of additional elements.

More information for figure 1

The image is an illustration depicting the formation of alloys. On the left, there's a frying pan with the label 'Fe', representing iron. In the middle, there's a yellow pencil with a pink eraser, typically made of other elements. On the right, there is a structural steel beam labeled 'Steel', illustrating a mixture of iron and carbon. This diagram highlights the transition from a pure base metal (Fe) to an alloy (Steel) with altered properties due to the addition of other elements.

[Generated by AI]

Making connections

In section S1.1.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) you learned about compounds and mixtures. Would you expect alloys to be an example of homogeneous or heterogeneous mixtures?

Alloys are homogeneous mixtures as they have a uniform appearance and composition throughout.

Note that in this section we have referred to alloys as mixtures instead of as compounds. We can distinguish between the two terms:

- Compound — atoms of different elements bonded together in a fixed ratio.
- Mixture — composed of two or more substances in no fixed ratio.

Why are alloys more correctly described as mixtures rather than as compounds?

Structure of alloys

Recall from section S2.3.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/metallic-bonding-and-structure-id-43464/) that the structure of elemental metals is a lattice of positive metal ions surrounded by delocalised valence electrons. The structure of alloys is similar to elemental metals, the key difference being that the added element will vary in size from the metal making up the base metal. There are two main types of alloys that form:

- Substitutional alloy - where the element added to the base metal simply replaces the metal ions in the lattice.
- Interstitial alloy - where the element added occupies vacant space in the metallic lattice of the base metal.

In both of these types of alloys, bonding is non-directional since the electrostatic force between cations and delocalised electrons still occurs in all directions. These two types of alloys are compared with that of an elemental metal below in **Figure 2.**

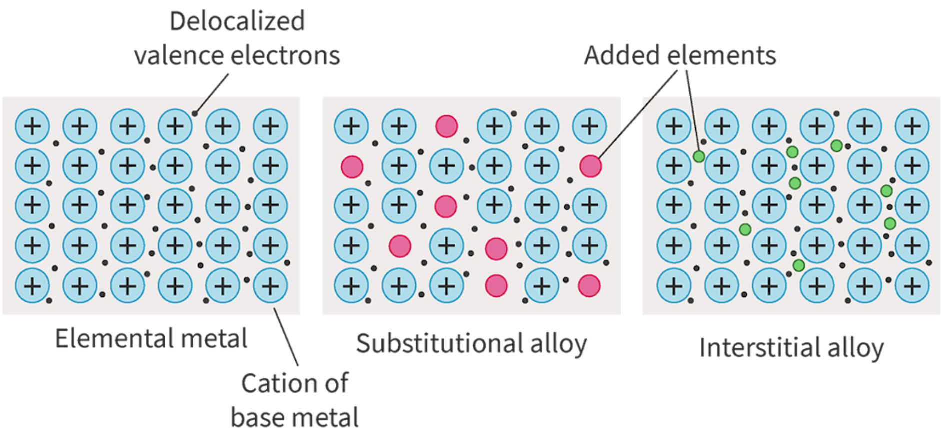


Figure 2. Comparison of the structure of an elemental metal, a substitutional alloy and an interstitial alloy.

More information for figure 2

The image is a diagram illustrating the comparison between the structures of an elemental metal, a substitutional alloy, and an interstitial alloy. On the left, the elemental metal structure is shown with blue circles representing metal cations and gray dots depicting delocalized electrons. In the center, the substitutional alloy is depicted with blue circles as the primary metal and red circles indicating substitute metal atoms, illustrating how different metal atoms can replace each other in the lattice. On the right, the interstitial alloy is illustrated with smaller green circles situated within the gaps of the metal lattice, indicating how smaller atoms occupy interstitial spaces. Labels describe each type of structure above the corresponding illustration.

[Generated by AI]

Study skills

You do need to describe alloys as mixtures of metals with other metals or non-metals and explain the properties of alloys in terms of non-directional bonding. The type of alloy (substitutional and interstitial) can be helpful in aiding understanding, but you are not required to recall these types in your examination.

In the structures of alloys in **Figure 2**, you can see how the base metal and the added element are not structurally arranged in a fixed ratio. Furthermore, since the bonding is between the ions in the lattice and the delocalised electrons in all directions, bonding is not between elements. This means there is no fixed ratio of elements bonded together. This is why alloys are more appropriately described as mixtures rather than compounds.

Physical properties of alloys

As you might expect, the introduction of different elements into the lattice of the base metal will influence the physical properties of the resultant material. Two physical properties that are most commonly affected through the process of alloying are:

- Malleability/hardness
- Melting point.

Let's look at each of these individually.

Malleability/hardness

In section S2.3.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/) we looked at how metals were malleable because layers in the lattice could easily slide past one another in response to an applied stress due to the non-directional bonding. What do you think will happen to the malleability of a metal when additional elements are added into the lattice structure to create an alloy? Would you expect layers to be able to slide past one another more easily, making the metal more malleable, or would it be more difficult for layers to slide past one another, making the alloy harder than the base metal?

The addition of elements with different sizes prevents the layers in the metallic lattice from sliding past one another easily. This causes the alloy to be less malleable and harder than the base metal. This is illustrated in **Video 1**.

Video 1. How alloys alter the properties of metals.

Melting point

Recall from section S2.3.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/strength-of-the-metallic-bond-id-43564/) that the stronger the electrostatic attraction between metal cations and delocalised electrons, the stronger the metallic bond and the higher the melting point. How might you expect the melting point of an alloy to differ from that of its base metal? How might the introduction of other cations into the metallic lattice influence the strength of the electrostatic attraction between cations and delocalised electrons? **Table 1** shows a comparison of the melting point temperatures for several elemental metals and alloys with those base metals.

Table 1. Comparison of the melting point temperatures between elemental metals and all

Elemental metal		Alloy	
Metal	**Melting point (°C)**	**Alloy**	**Approxima melting point**
Copper	1085	Bronze (copper with tin)	913
		Brass (copper with zinc)	930

Elemental metal		Alloy	
Metal	Melting point (°C)	Alloy	Approxima melting point
Iron	1538	Steel (iron with carbon)	1515
		Stainless steel (iron with chromium, and nickel)	1425
Aluminium	660.3	Aluminium alloy (aluminium with copper, manganese, nickel, zinc)	565
Tin	231.9	Solder (50% tin, 50% lead)	215
Lead	327.5		

Look at the melting points of tin and lead in **Table 1**. Now look at the approximate melting point for solder, an alloy made of tin and lead. What do you notice about the melting point for these two metals when they are combined as an alloy?

Notice that it is lower. What about the other alloys in **Table 1**?

The addition of elements into the lattice structure of a base metal will influence the strength of the metallic bond. For many alloys, this lattice structure with varying cation size weakens the electrostatic attraction between the cations and the delocalised electrons. This causes many alloys to have lower melting temperatures than the base metal due to this weaker metallic bond.

You may also have noticed in **Table 1** that the melting point for alloys is listed as an approximate temperature. This is because the melting point will change depending on the amount of alloying element added. Chemists studying alloys regularly make use of a phase diagram which helps to track the melting point of alloys as the percent composition is altered. To construct a phase diagram, scientists would study what temperature the alloy of each percentage composition would:

1. show signs of melting
2. complete melting.

Figure 3 is an example of a phase diagram for a tin-lead alloy, commonly known as solder.

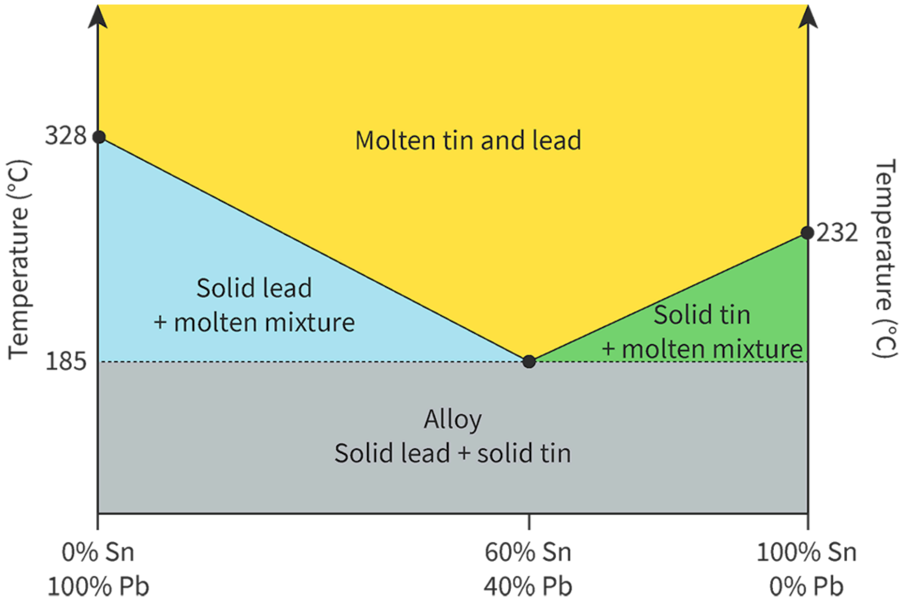


Figure 3. Phase diagram for tin-lead alloys.

More information for figure 3

Practical skills

Tools 1: Experimental techniques — Addressing safety

Inquiry 1: Exploring and designing — Exploring

Tin and lead are both examples of metals with low melting points. This means that you can melt these metals to create alloys easily using hot plates commonly found in a high school chemistry laboratory. Can you think of ways you could safely design an experiment to create a tin-lead alloy of varying percent composition? What equipment would you need? How might you study the melting temperature of your alloy either quantitatively or qualitatively? What are the safety considerations you would need to take to handle a substance like lead that has known toxicity concerns?

Common examples of alloys

Many other physical properties of alloys vary from their base metals, but they tend to be unique to specific alloys as opposed to all alloys. Let's look at some common examples of alloys and their unique physical properties in **Table 2**.

Table 2. Common examples of alloys and their unique physical properties.

Alloy	Constituent elements	Physical properties	Uses
Steel	Fe + C	High strength, very hard material	Constructic materials
Stainless steel	Steel + Cr, Ni	Resistant to corrosion Metal does not rust as easily	Kitchenware surfaces that r to be cleaned r
Bronze	Cu + Sn	Hard material, low melting point	Weapons and
Brass	Cu + Zn	Resistant to corrosion	Musical instrun gears, hinges valves
Aluminium alloy	Al + Cu, Mn, Ni, Zn	High strength, resistant to corrosion Low density (due to low density of Al)	Aerospace manufacturing gas cylinders scuba divir
Solder	Sn + Pb	Low melting point More conductive than Pb but less conductive than Sn	Connectin electrical wires plumbing pie
Amalgam	Hg + Ag, Sn, Cu	Hardens quickly upon formation of alloy Resistant to corrosion	Dental fillin
Sterling silver	Ag + Cu	Strong and hard Resistant to tarnish	Victorian era c sets and jewe
White gold	Au + Pd, Ni, Zn	Hard and strong Light grey colour	Jewellery
Rose gold	Au + Cu	Hard and strong. Red hue in colour	Jewellery

You may have noticed that many of the alloys in **Table 2** have resistance to corrosion as a physical property. Corrosion is the deterioration of a material on its surface as it interacts with substances in the surroundings. For metals, corrosion is typically the formation of brittle metal oxides on the surface due to the process of oxidation. You will learn more about the process of oxidation in section R3.2.1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/oxidation-and-reduction-id-43487/).

Study skills

Note that you do need to be able to explain why the properties of alloys differ from their base metals. However, you are not required to know specific examples of alloys. These examples simply aid in the understanding of these differing properties.

In this next activity you will research and become an 'expert' on one specific alloy that you will teach your group all about.

Activity

- **IB learner profile attribute:** Risk-taker
- **Approaches to learning:** Research skills — Using search engines and libraries effectively
- **Time required to complete activity:** 45—60 minutes
- **Activity type:** Individual activity for #1, groups of 4—5 for #2

Alloy research jigsaw activity

1. **Home groups**
   In groups of 4—5, look through the following list of alloys. Each group member will select one alloy to research and become an 'expert' on.

	Types of Alloys
Alloys	• Hiduminium (aluminium based alloy) • Wood's metal (bismuth based alloy) • Nichrome (nickel based alloy) • Pewter (tin based alloy) Galinstan (gallium based alloy)

2. **Expert groups**
   Create a new expert group with individuals in the class that have chosen the same alloy as you. In this expert group you will research the specified alloy as a team (approximately 10 minutes). Research the following:

   (a) Composition of alloy.
   (b) The approximate melting point of the alloy.
   (c) Unique properties this alloy has in comparison to its base metal.

(d) Examples of where this alloy is used.

Sketch a diagram representing the structure of the alloy you have research.

As expert groups, evaluate information sources you are using for accuracy, bias, credibility and relevance. You will be sharing your research findings with your home group. You may want to take notes during your research to help you with this.

3. **Return to home groups**
   Returning to your home group, each member of the group will share research findings of their alloy. You may consider using a slideshow, chart paper, whiteboard, etc. to share your research findings.

Polymers

Learning outcomes

By the end of this section you should be able to:

- Describe what polymers are.
- Describe the common properties of plastics in terms of their structure.

In our discussion of ionic, covalent, metallic bonding we have encountered several different solids. We have looked at ionic minerals, ionic salts, molecular solids, covalent network structures, metals and alloys. There is one type of solid material that is very commonly used globally that we have not mentioned so far. Can you think of what that might be? Do you need a hint? Look around the room you are in now. What substances are objects made of? Metals, wood, glass...

Another common type of material that is used to make a very wide range of objects are polymers. Usually, plastics are the first thing to come to mind when people hear the word polymers, but plastics are not the only type of polymers, as you will see throughout this and the following sections in the subtopic.

But what are polymers? What type of atoms create polymers? What is the structure and bonding of a polymer? Why does it have properties that societies find so useful? Why don't they all have the same properties? Why are some of these polymers a global environmental concern?

The term polymer is an Ancient Greek word meaning 'many parts'. Polymers are large molecules, or macromolecules, made from repeating subunits called monomers. These monomers are small molecules that make up the building blocks of a polymer. **Figure 1** below uses building blocks to illustrate the concept of a polymer having many parts.

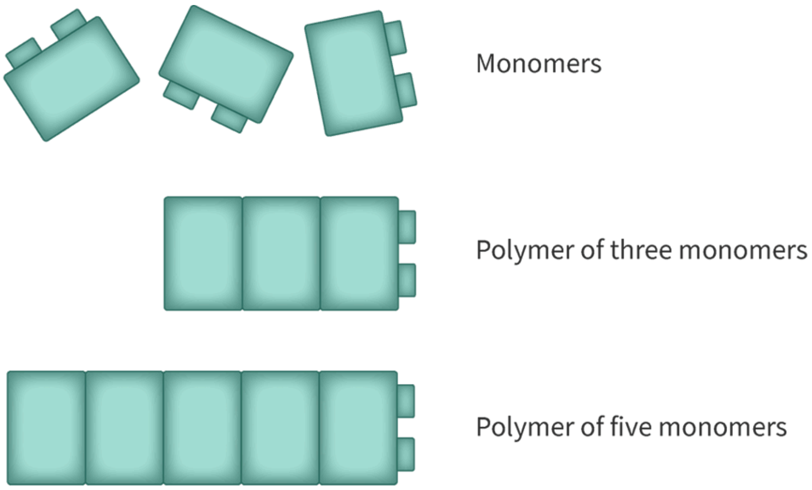


Figure 1. Polymers are macromolecules made up of monomers as the building blocks.

More information for figure 1

The image illustrates the structure of a polymer using a block model, conveying the concept of macromolecules composed of repeating subunits called monomers. The polymer is represented by multiple small block-like structures arranged in sequences to form a larger assembly, visually likening each block to a monomer within the polymer chain. The blocks are oriented in a grid-like fashion, showcasing multiple rows of connected blocks to emphasize their repetitive nature. This arrangement symbolizes how polymers are made from repeating units that build complex macromolecules.

[Generated by AI]

Recall from section S2.2.7 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/covalent-network-structures-id-43593/) that carbon can form a variety of different covalent network structures, called allotropes in addition to the large variety of small molecules. This is largely due in part to the four valence electrons allowing the formation of four covalent bonds. Carbon is also the backbone of most polymer structures. **Figure 2** shows an example of a polymer called polypropene (also known as polypropylene), made up only of carbon and hydrogen. Notice how it is a large molecule, hence, macromolecule, with many parts.

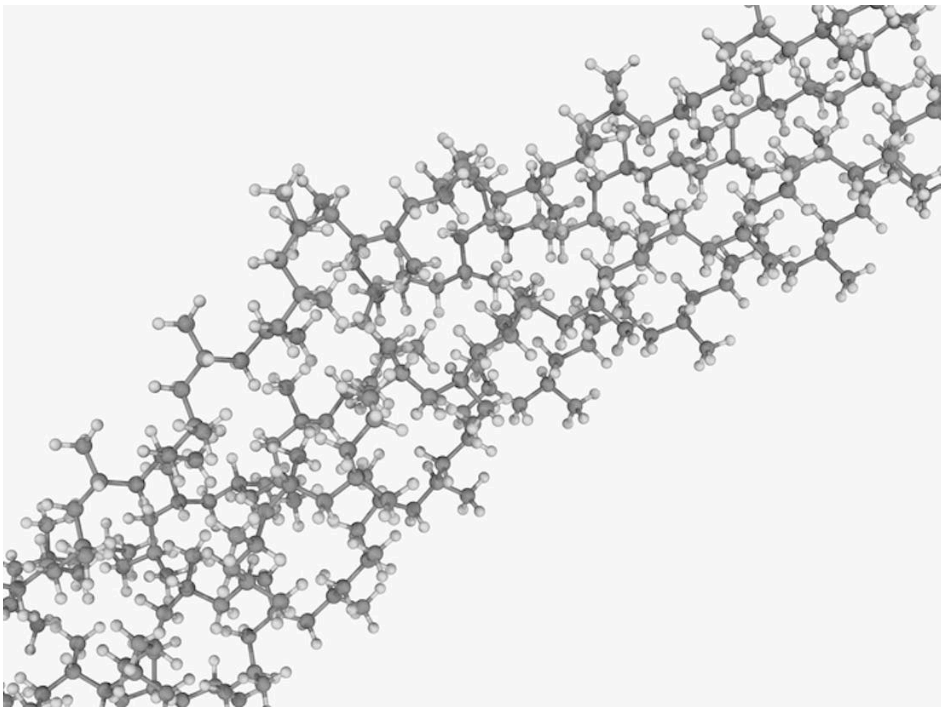


Figure 2. Example of a polymer structure, polypropene.

Credit: LAGUNA DESIGN, Getty Images

More information for figure 2

The image illustrates the molecular structure of polypropene, a synthetic polymer composed of repeating units derived from propylene monomers. The structure shows multiple chains of carbon and hydrogen atoms interconnected to form a larger macromolecule. This visual gives a detailed account of how the individual carbon atoms bond together, showcasing its complexity and the repeating pattern inherent in polymers. The structure is three-dimensional and complex, with each unit connecting to others in a network-like configuration.

[Generated by AI]

There are other ways that we can represent the polymer polypropene. In **Figure 2** we can see that it is a large molecule with many parts, but it is more difficult to see what the repeating unit actually is. Structural formulas are one way to show how atoms are bonded in the polymer structure that clearly shows the repeating unit. **Figure 3** shows a few different ways that the structural formula can be presented.

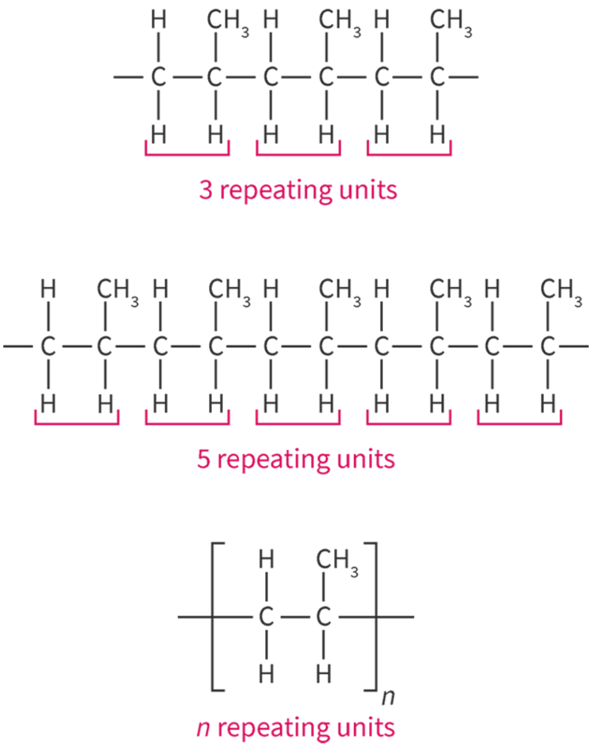


Figure 3. Structural formulas for polypropene.

More information for figure 3

The image illustrates different structural formulas of polypropene. It shows three variations: the first with three repeating units, the second with five, and the third with "n" repeating units, indicating continuation. Each formula is depicted with covalent bonds at the ends of the polymer chains left incomplete, suggesting the polymer's potential to extend indefinitely in either direction. This visualization helps in identifying the repeating unit of the polymer and understanding the polymer structure more clearly.

[Generated by AI]

Notice from **Figure 3** how the structural formula makes it much easier to identify the repeating unit of the polymer. The first example shows three repeating units, the second one shows five repeating units, and the third one shows *n* repeating units. The variable *n* is often used to denote that there can be any number of repeating units, 100, 1 million and usually even more! You will notice that at the end of the polymer chains the covalent bonds drawn are left incomplete to indicate the polymer can continue to repeat in either direction.

Plastic

Plastic is a category of polymer that can be moulded or extruded into a desired shape. Plastics are a wide range of materials used for consumer products, medical products, packaging, textiles, electronics, transportation, construction and more. **Figure 4** shows some common examples of plastic polymers and their structural formulas.

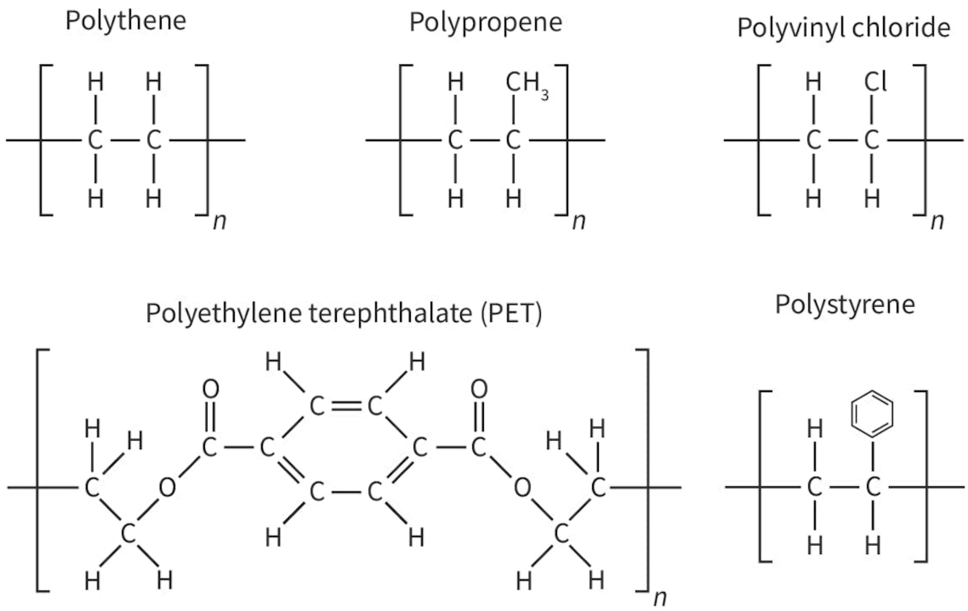


Figure 4. Common plastic polymers.

More information for figure 4

The image displays the structural formulas of six common plastic polymers: polythene, polypropene, polyvinyl chloride, polyethylene terephthalate (PET), and polystyrene.

1. **Polythene:** Shows a repeated unit of two carbon atoms (C) each bonded to two hydrogen atoms (H), with linear bonds extending to the sides indicating continuation.

2. **Polypropene:** Similar to polythene, but one hydrogen in the repeated unit is replaced with a methyl group (CH_3) on the second carbon.

3. **Polyvinyl chloride:** Formed by a repeated unit of two carbon atoms where one hydrogen is replaced by a chlorine atom (Cl) on the second carbon.

4. **Polyethylene terephthalate (PET):** Displays a complex chain structure with ester linkages (—COO—) and a phenyl ring in the middle.

5. **Polystyrene:** Consists of a repeated unit of two carbon atoms, with one hydrogen replaced by a phenyl group (a benzene ring with alternating double bonds) on the second carbon.

Each polymer is labeled above its structural formula. The repeated units are enclosed in square brackets with a subscript 'n' to indicate polymerization.

[Generated by AI]

Some of the main properties of plastics include:

- Flexible
- Durable
- Mould easily into desired shapes
- Lightweight
- Good electrical and thermal insulators

The polymer polypropene we looked at in **Figure 2** and **Figure 3** is also a plastic polymer. What do you think it is about its structure that allows it to have these properties? In subtopic S2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/) you looked at molecular and covalent network structures. What was the large difference in structure that caused covalent network structures to be hard, rigid and have high melting points in comparison to molecular solids?

Covalent network solids formed three-dimensional lattice structures held together by covalent bonds whereas molecular solids involve individual atoms covalently bonded within molecules that are held together by intermolecular forces. Polymer structures are somewhere in between these two. They involve very long chains of molecules, but the individual polymer chains are held together by intermolecular forces. The intermolecular forces holding separate polymer chains together is similar to how intermolecular forces hold separate sheets together within graphite. A comparison of the structures of a molecular solids, covalent network and polymer are shown in **Interactive 1**. Click through each object to zoom in on the structure.

Interactive 1. Interactions Between Polypropene Polymer Chains.

More information for interactive 1

An interactive collage features three images of materials formed by covalent bonding. Each image has a plus sign, indicating an interactive hotspot. Selecting these hotspots reveals the description and chemical structure of the materials.
Left: Dry ice
Image: A white block of dry ice on a glass container, with white smoke coming out.
Structure: A cubic structure with CO_2 molecules occupying the edges and face centers of the cube.
Description on clicking the hotspot: Covalent bonds within each CO_2 molecule. Molecules held together by intermolecular forces in solid.
Middle: Polypropylene plastic bottle
Image: Three colored polypropylene plastic bottles displayed together.
Structure: The chemical structure features three rows of polypropylene. Polypropylene is made up of repeating propene (C_3H_6) units. Each row features a long carbon chain with alternating hydrogen (H) and methyl (CH_3) attached on the top and hydrogen (H) atoms attached to the bottom.
Description on clicking the hotspot: Covalent bonds within each polymer chain. Polymer chains held together by intermolecular forces.
Right: Diamond
Image: A shiny, reflective diamond.
Structure: A 3D network structure of carbon atoms arranged in four layers. Each carbon atom is tetrahedrally bonded to four other carbon atoms.
Description on clicking the hotspot: Held together by covalent bonds.

Flexibility and durability

You can see in **Interactive 1** that polymers have weak intermolecular forces that hold polymer chains together. These intermolecular forces are what gives plastic polymers flexibility and durability. Polymer chains can slide past one another but the polymer itself remains intact. Since flexibility is the ability of a material to be bent without breaking, the weaker the intermolecular forces between polymer chains, the more flexible a plastic is. Conversely, the stronger the intermolecular forces between polymer chains, the more rigid a plastic is.

Durability is the ability of a material to be functional over extended periods of time. Polymer chains are made up of a very large number of covalent bonds that do not break apart easily. This results in a long-lasting material that is durable. While the durability of plastics is useful can you think of why they might be of concern? Since the large number of covalent bonds do not break apart easily, they are difficult to break down. Once a plastic is made it is very difficult to get rid of. This is a global concern as plastics accumulate in landfills all over the world.

Nature of Science

Aspect: Global impact of science

Did you know there are newer plastics out there being developed to help reduce waste? These plastics are called bioplastics made up of starches, cellulose or biopolymers, obtained from renewable resources such as food waste. When these plastics are disposed of in the landfill, ordinary bacteria can break down the starch. Learn about the production, challenges and future of bioplastics in this article (https://www.sciencedirect.com/science/article/abs/pii/S2214799320301107).

Ability to mould into a desired shape

Since polymers are formed from smaller molecules that make up the repeating subunits called monomers, they can be moulded into shape during the formation of the polymer as monomers react together. In addition, since polymer chains are held together by weak intermolecular forces, many polymer plastics can be melted and reshaped to form new materials. You may have encountered this if you have placed something plastic too close to a heat source like a stove. A plastic container for example will melt and reshape when placed too close to the stove.

Lightweight

Considering the structure of polymers and the types of atoms that make up many plastic polymers, why do you think plastics are lightweight? Polymer chains are held together by weak intermolecular forces in polymers. Since this interaction is much weaker than covalent, metallic or ionic bonds, polymers do not pack very closely together in comparison to ionic, covalent networks and metallic lattices. This leads to polymers having a lower density than solids with an ionic, covalent network, or metallic lattice. Additionally, most polymer plastics are made up of long chains of carbon atoms with primarily hydrogen, oxygen, nitrogen and chlorine covalently bonded to these carbon chains. Carbon has a relatively low atomic mass, further contributing to the low density, lightweight nature of plastics.

Insulating

You learned about the electrical and thermal conductivity of metals in section S2.3.1b (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-properties-and-application-of-metals-id-43563/). Have you noticed how plastic is often used to coat electrical wires and the handles on cookware? If you look at the plastic polymer structures in **Figure 4**, can you predict why plastic polymers are poor electrical and thermal conductors? In all of the plastic polymer examples, there are no examples where electrons can delocalise throughout the polymer chains. This makes plastic polymers poor electrical and thermal conductors, which therefore make them good electrical and thermal insulators.

International Mindedness

The development of the polymer and plastics industry was actually motivated from trying to preserve natural resources.

1. The first synthetic polymer was produced by John Wesley Hyatt. He was looking to produce a material that could substitute the use of ivory in billiard balls. As the game billiards increased in popularity, natural ivory from elephant tusks became very scarce. The plastic produced by John Hyatt was developed to protect the natural world from human desires.
2. During World War II, natural resources became very scarce. New synthetic polymers were used to preserve natural resources and provide substitutes for military items.

The irony is now the production, consumption and disposal of plastic pose one of the world's largest threats. Responsible consumption and production is #12 of the UN's sustainable development goals (https://sdgs.un.org/goals). We all have a responsibility at the local, national and international level to evaluate the sustainability of production, reduce our consumption and re-evaluate the disposal of plastics.

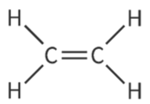
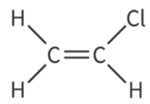
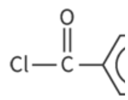
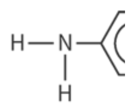
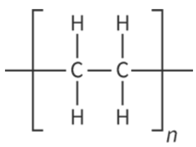
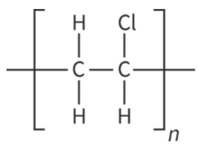
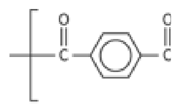
Properties of polymers and intermolecular interactions

The intermolecular interactions between polymer chains are correlated with the observed physical properties. Three commonly encountered plastic polymers are:

- Polyethylene (also known as polyethene)
- Polyvinylchloride (also known as PVC or polychloroethene)
- Kevlar

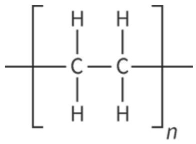
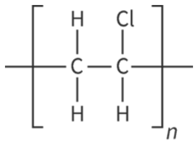
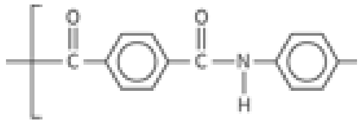
Table 1 shows the structures for the polymer and monomer for these three polymers. Look at each of them and identify what intermolecular forces (London dispersion forces, dipole-dipole forces, and hydrogen bonding) you would expect to exist between polymer chains. Which would you expect to be the strongest plastic polymer by looking at the structure of the monomers and polymers? Which would you expect to be the most flexible plastic?

1. Structural formula of three polymers and the monomers they are are prc

Polyethylene	Polyvinylchloride	Ke
 More information	 More information	 a 
 More information		 Mo

As these three polymers each have a different dominant intermolecular force between polymer chains, plastics made from these polymers will have differing properties. **Table 2** provides a summary of these intermolecular forces, properties and an example of where these polymers are used.

Table 2. Intermolecular forces and properties of three commonly encountered plastic poly

Polymer	Polyethylene	Polyvinylchloride	Kevlar
Polymer structure			
Intermolecular forces between polymer chains	London dispersion forces	London dispersion forces and dipole-dipole forces	London dispersion forces hydrogen bonding
Properties	Flexible	Rigid	Very strong and tough
Example of use	Plastic bags	Plumbing	Bulletproof vests

Natural polymers

So far, we have considered only synthetic polymers. Synthetic materials are those made by humans that cannot be found in nature. Monomers for synthetic polymers are usually derived from fossil fuels. Polymers have been around long before we humans have been synthesising them! Polymers found in nature are referred to as natural polymers. Here are some examples of natural polymers:

- Sugars such as cellulose found in wood and plants, and starch found in foods such as rice, potatoes and wheat.
- Proteins making up your skin, hair, muscles and nails are polymers made out of amino acids as the repeating units.
- DNA making up your genetic code is made up of two polymer strands that interact with one another through hydrogen bonding forming a double helix.
- Spider silk made from a polymer that has structural similarities to Kevlar.

Interactive 2 shows the structures of some of these natural polymers. Scroll through to see them.

Interactive 2. Examples of Natural Polymers.

More information for interactive 2

Making connections

Synthetic polymers are typically produced using starting materials from petroleum oil. The mass production of plastics globally has increased the demand on fossil fuels. To reduce the demand on fossil fuels but still produce plastic, a group of polymers known as biodegradable polymers has been created. Here are some examples of these structures.

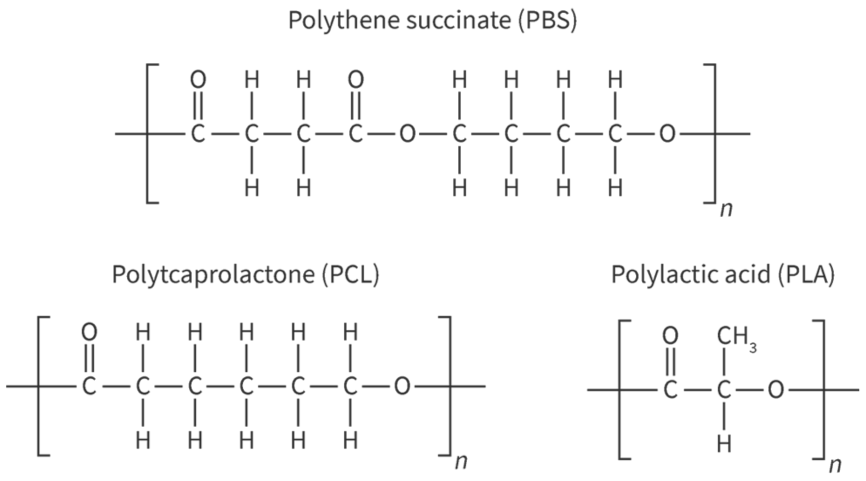


Figure 5. Examples of biodegradable polymers.

More information for figure 5

What structural features do these polymers have in common? Are they more structurally similar to synthetic or natural polymers? What structural features make them biodegradable?

Before completing this activity you will need to collect all the plastic waste you generate or encounter in a 24 hour time period. You will then work with a group to categorise these plastic polymers based on your physical observations.

Activity

- **IB learner profile attribute**: Caring
- **Approaches to learning**: Communication skills — Practising active listening skills
- **Time required to complete activity**: 30 minutes
- **Activity type**: Group and whole class

Individually

Over the next 24 hours collect all of the plastic waste or products that you encounter and bring them to class. You should aim to bring in a minimum of five plastic objects.

In groups of 4—5

1. Gather all of your plastic objects and inspect them.
2. Create categories of the plastics using your physical observations of the similarities and differences of the objects. Come up with a title for each category.
3. Create a list of five advantages and five disadvantages of plastic polymers.
4. Research a local recycling program (could be from your school, municipal, district, etc.). Identify what types of plastics are recycled and if any information is provided about what plastic polymers are recycled.
5. Identify an alternative non-plastic option for each of the items each group member brought in.

As a class

1. In a class discussion, each group will share their categories for their plastic objects providing justification for how they decided upon the placement of their objects into the categories.
2. Create a class list of advantages and disadvantages of plastic polymers.
3. Discuss what are individual actions each of us can take to work towards meeting the UN's sustainable development goals (https://sdgs.un.org/goals) #12 responsible consumption and production.

S2.4 From models to materials

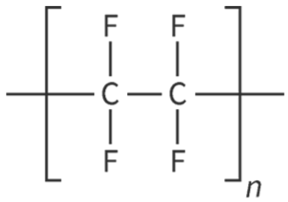
Addition polymers

Learning outcomes

By the end of this section you should be able to:

- Represent the repeating unit of an addition polymer from given monomer structures.
- Deduce the monomer for an addition polymer given the polymer structure.

Teflon™ is a material commonly used in non-stick cookware. You might have a frying pan in your kitchen that has a teflon coating on the cooking surface. Teflon is made of the polymer tetrafluoropolyethene (PTFE). This polymer has a relatively high melting point (327 °C), it does not interact with water and is very unreactive with most substances. These properties make it ideal for use in cookware. The structure of tetrafluoroethene is shown in **Figure 1**.

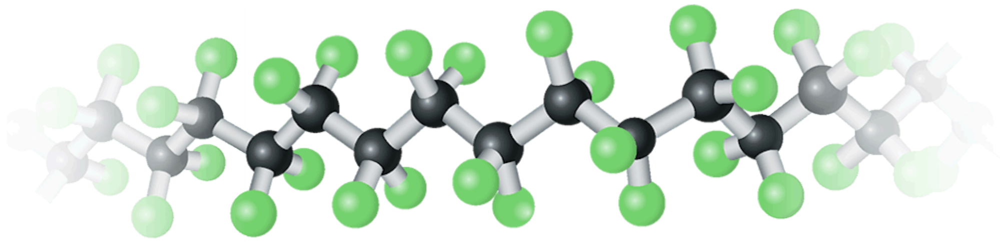


Polymer

More information

The image is a structural diagram of the chemical compound tetrafluoroethene (TFE), which is a monomer used to produce the polymer polytetrafluoroethene (PTFE, commonly known as Teflon™). The diagram shows two carbon atoms (C) double-bonded to each other at the center. Each carbon atom is also bonded to two fluorine atoms (F), positioned at the top and bottom. This configuration creates a linear, symmetrical structure. The chemical formula is often represented as C2F4. TFE is a colorless gas under standard conditions, and the inclusion of fluorine atoms makes it highly resistant to chemical reactions, contributing to its non-stick properties.

[Generated by AI]



Ball and stick model

Figure 1. Structure of polytetrafluoroethene.

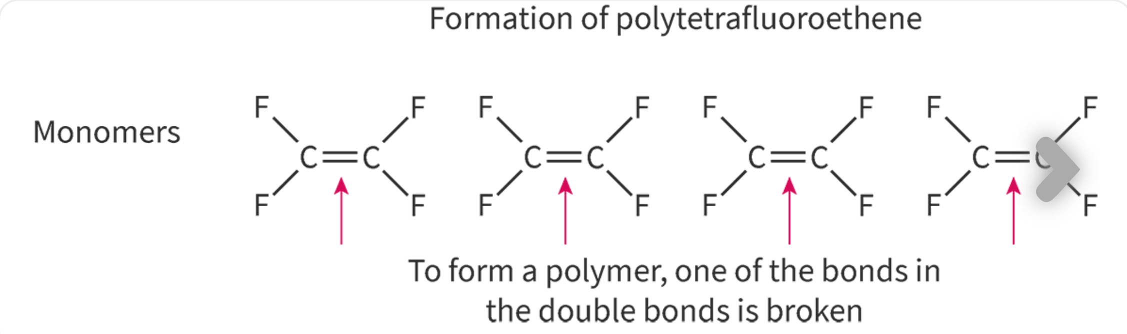
More information for figure 1

The image is a diagram illustrating the molecular structure of polytetrafluoroethene (PTFE). It depicts a repeating chain of carbon atoms, each bonded to two fluorine atoms. The diagram is arranged linearly, indicating a straight chain polymer. The carbon atoms appear in black, while the fluorine atoms are shown in green. The bonds between the atoms are represented by lines connecting individual spheres, which signify atoms. This diagram helps visualize the molecular construction of PTFE, emphasizing the alternating pattern and linear structure typical of this polymer.

[Generated by AI]

But how is a polymer such as PTFE formed? What is the structure of the monomer molecule building blocks used to form PTFE? What happens to the monomer in order to form a polymer like this?

Recall from section S2.4.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/polymers-id-43845/) that polymers are large macromolecules made from repeating subunits called monomers. These monomers act as the building blocks to form polymers. One method used to form a polymer is through a process called addition polymerisation. Addition polymerisation is the process of forming a molecule by simply adding monomers together. For this to occur, the monomer must have a double bond between two carbon atoms in its structure. When the polymer is formed, one of the bonds in the double bond is broken in each monomer allowing a single bond to form with other monomers. This process is illustrated in **Interactive 1** for the formation of the polymer polytetrafluoroethene. Press the arrow to view the slides.



Interactive 1. Formation of Polytetrafluoroethene.

More information for interactive 1

A slideshow with four slides displays the formation of polytetrafluoroethene.
Slide 1: Each monomer consists of two carbon atoms double-bonded to each other, with each carbon atom bonded to two fluorine atoms. Red arrows point toward the double bonds between carbon atoms to highlight the breaking of one bond. Onscreen text: To form a polymer, one of the bonds in double bonds is broken.
Slide 2: Each monomer now consists of one single bond between carbon atoms, with each carbon bonded to two fluorine atoms. Arrows from the carbon atom of one monomer to the carbon atom of the next monomer are drawn. The onscreen text below reads as follows: These carbon atoms are not stable. These monomers bond with adjacent monomers to become more stable.
Slide 3: The image demonstrates the polymerisation process where the double bond between carbon atoms is broken, forming a continuous chain of carbon atoms with fluorine atoms attached. Onscreen text: These newly formed bonds between monomers form a polymer. Since monomers were simply added together these polymers are called addition polymers.
Slide 4: The polymerisation process of tetrafluoroethene (C subscript 2 F subscript 4) to form polytetrafluoroethene. On the left side, the monomer structure shows two carbon atoms double-bonded to each other, with each carbon atom individually bonded to two fluorine atoms. The notation n indicates multiple monomer units. An arrow points to the polymer structure on the right side, where the repeating unit consists of single bonds between carbon atoms, each bonded to two fluorine atoms. The polymer chain is enclosed in brackets with the n notation on the outside, signifying numerous repeating units.
The interactive demonstrates the polymerisation reaction of tetrafluoroethene.

Notice in **Interactive 1** how the polymer formed is named after the monomer with the prefix poly- added in front. This is how all addition polymers are named. Note that many of the plastics we looked at in section S2.4.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/polymers-id-43845/) were examples of addition polymers. Looking at the plastic addition polymers shown in **Figure 2**, see whether you can name and draw the structures for each of their monomers.

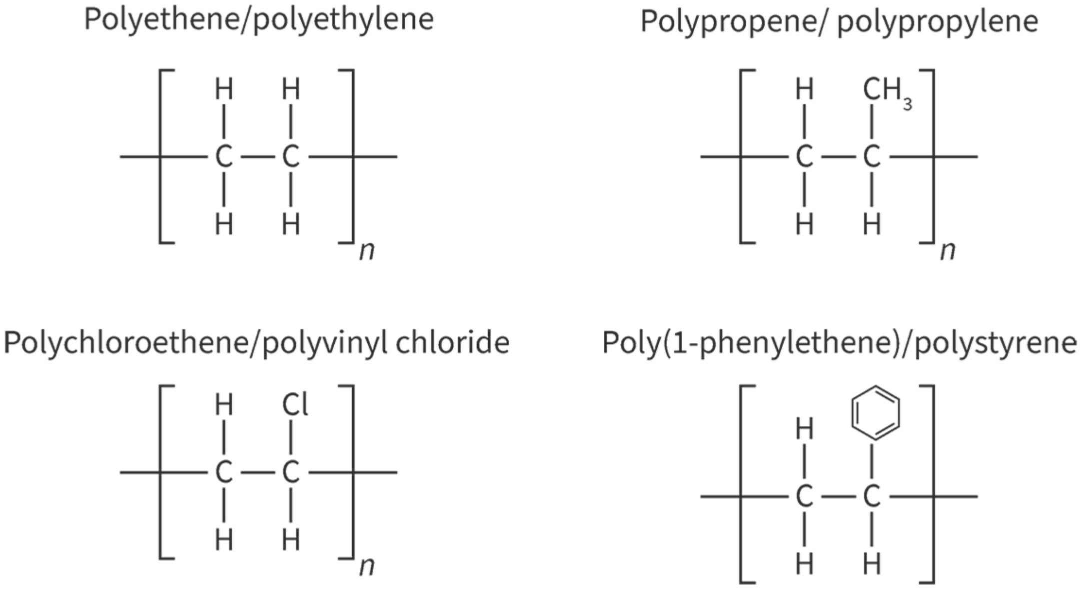


Figure 2. Common plastic addition polymers.

More information for figure 2

This diagram presents the chemical structures and names of four common plastic addition polymers. The polymers are displayed in a grid-like format. From top to bottom, left to right: the first is Polyethylene, depicted with a chain of repeating ethylene units (CH2-CH2); next is Polypropylene, shown with repeating propylene units (CH2-CH(CH3)); followed by Polyvinyl Chloride, represented with repeating vinyl chloride units (CH2-CHCl); and lastly, Polystyrene, shown with a chain of styrene units (CH2-CH(C6H5)). Each polymer's structural formula is labeled accordingly, highlighting the differences in their monomer units. This demonstrates the variety in structure among addition polymers.

[Generated by AI]

Nature of Science

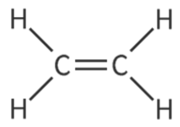
Aspect: Science as a shared endeavour

To name compounds a set of rules is set by the International Union of Pure and Applied Chemistry (IUPAC (https://iupac.org/)). These rules allow scientists all over the world to communicate information about the substances they are studying. You may notice however that there is some variation in name for the same polymer labelled in commercial products. For example, polytetrafluoroethene would be the IUPAC polymer name but polytetafluoroethylene is used commercially. Both refer to the same polymer.

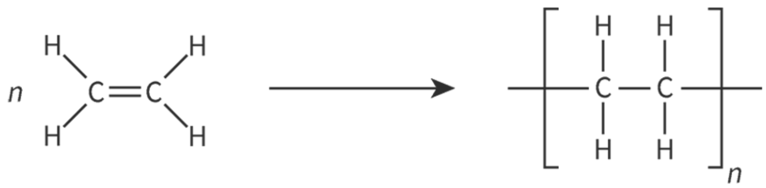
Worked example 1

What is the name and the structure of the addition polymer formed from ethene, C_2H_4?

The molecule C_2H_4 has the following structure:



When the polymer is formed, one of the bonds in the double bonds breaks to join monomers together to form the polymer with the following structure:



Since the polymer was made from the monomer ethene, the name would be polyethene.

Worked example 2

The following polymer is known as polyvinylidene dichloride. Identify the repeating unit and deduce the structure of the monomer for the following polymer.

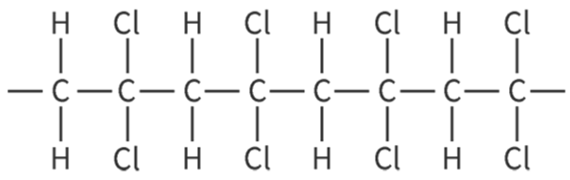


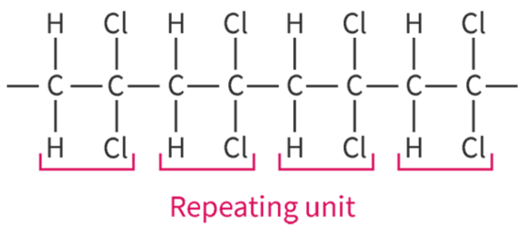
Figure 3. Structural formula of polyvinylidene dichloride.

More information for figure 3

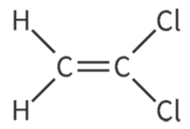
The image depicts the structural formula of polyvinylidene dichloride. It shows repeating units characterized by a vinylidene group bonded to a dichloride group. The chemical structure features a polymer backbone formed by carbon atoms, each carbon atom alternately double-bonded to another carbon atom while also sharing single bonds with two hydrogen atoms. Each repeating unit within the polymer chain includes two chlorine atoms attached to carbon atoms, contributing to the overall molecular structure of polyvinylidene dichloride. The formula is representative of how the polymer's monomer units are linked chemically to form its macromolecular configuration.

[Generated by AI]

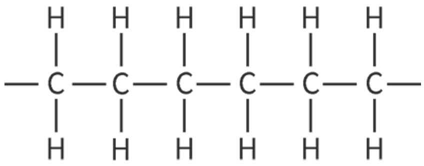
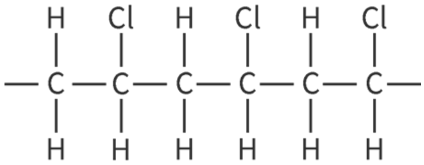
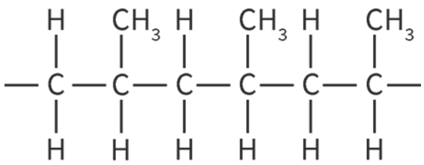
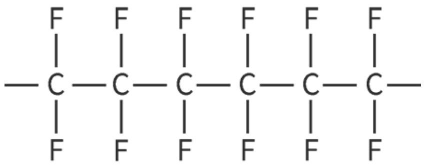
Looking at the polymer the repeating unit would be:



This would indicate that the monomer has two carbon atoms, two hydrogen atoms and two chlorine atoms in its structure. Now in order to form a polymer from the monomer a bond between carbon atoms would need to break. This means the monomer has a double bond between carbon atoms. Therefore, the structure of the monomer would be:



In **Interactive 2** drag and drop the names of monomers to match them to the correct polymer structure.

Polymer structure	Name	Monomer
		
		
		
		

polypropene

polytetra-fluoroethene

polyethene

poly-chloroethene

tetra-fluoroethene C_2F_4

ethene C_2H_4

propene C_3H_6

chloroethene C_2H_3Cl

Check

Interactive 2. Polymers and Their Monomers.

More information for interactive 2

An interactive table illustrates polymer structures where users need to drag and drop the corresponding names and monomers for the given structures.
The provided polymer structures are described as follows:
Structure 1:
Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-bonded to hydrogen atoms at the top and bottom.
Structure 2:
Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-

Kog

bonded to a hydrogen atom at the bottom. At the top, hydrogen and chlorine atoms are bonded alternately to the carbon atoms.
Structure 3:
Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-bonded to a hydrogen atom at the bottom. At the top, hydrogen atoms and methyl groups are bonded alternately to the carbon atoms.
Structure 4:
Six carbon atoms are arranged in a straight line, connected by single bonds. An extra bond extends from the first and the last carbon atoms. Each carbon atom is single-bonded to fluorine atoms at the top and bottom.
The drag-and-drop options are provided as follows:
Names:
Polypropene
Polytetrafluoroethene
Polyethene
Polychloroethene
Monomers:
Tetrafluoroethene C_2F_4
Ethene C_2H_4
Propene C_3H_6
Chloroethene C_2H_3Cl
This interactivity enables users to identify the names and monomers of various polymer structures.
Read below for the solution:
Structure 1:
Name: Polyethene
Monomer: Ethene C_2H_4
Structure 2:
Name: Polychloroethene
Monomer: Chloroethene C_2H_3Cl
Structure 3:
Name: Polypropene
Monomer: Propene C_3H_6
Structure 4:
Name: Polytetrafluoroethene
Monomer: Tetrafluoroethene C_2F_4

Creativity, activity, service

Strand: Service
Learning outcomes:

- Recognise and consider the ethics of choices and actions.
- Demonstrate engagement with issues of global significance.

Kog

Polymers are often chosen for their durability and longevity. However, it is these properties that result in plastics causing long-term problems (https://www.worldwildlife.org/stories/the-problem-with-plastic-in-nature-and-what-you-can-do-to-help) in the natural environment. According to the WWF, a plastic straw can take up to 200 years to break down.

Reducing the use of single-use plastic is one way in which we can all make a difference. Another way is to raise awareness of the importance of reducing plastic waste. Read about more ways to reduce the use of plastics in your school here (https://www.sustainablemerton.org/post/what-can-schools-do-to-reduce-plastic-pollution).

The next activity will allow you to identify the names and structures of varying addition polymers and/or their monomers, along with researching the uses for each polymer.

Activity

- **IB learner profile attribute**: Inquirer
- **Approaches to learning**: Research skills — Using search engines and libraries effectively
- **Time required to complete activity**: 20—30 minutes
- **Activity type**: Individual activity

Complete the following worksheet.

Worksheet (https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

In this worksheet you will need to complete the following:

- Deduce the name and structure of a polymer given the monomer.
- Deduce the name and structure of the monomer given the polymer.
- Research any practical applications and uses each polymer has.
- Check your worksheet answers with a classmate or when you research the practical uses of polymers.

S2.4 From models to materials

Condensation polymers (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Represent the repeating unit of polyamides and polyesters from given monomer structures.
- Deduce the monomers for a condensation polymer given the polymer structure.

Take a quick look at the tag inside your clothing. What material is listed on the tag? Is it made up of one material or a combination? Many clothing threads are made of polymers. Some are natural polymers such as cotton, wool or rayon, and some are synthetic polymers like polyester, polyamide (also known as nylon) or spandex. But how are these polymers formed? What is the structure of their monomers? How are these polymers similar or different to addition polymers we looked at in section S2.4.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/)?

International Mindedness

As producing synthetic polymers fibres has become more inexpensive, the fashion industry has grown tremendously, particularly in the direction of fast fashion. Fast fashion is the rapid design, production and marketing of garments to quickly keep up with current trends using low quality materials. Some of the major issues of fast fashion are:

1. Using fossil fuels for the production of polymer fibres.
2. Clothing is being discarded rapidly due to poor quality or new trends emerging.
3. Ethical concerns about the treatment of garment workers.

The UN's sustainable development goal (https://sdgs.un.org/goals) #12 is focused on responsible consumption and production. The fast fashion industry is certainly one to be considered here. When purchasing your next garment, consider your need, how long you intend to keep it, and its quality.

In section S2.4.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/) you learned about addition polymers where monomers containing a double bond were able to add together without forming any byproducts, only the polymer. Most clothing fibres are instead condensation polymers. A condensation polymer is formed when two monomers react, releasing a small molecule, such as water, as a byproduct. Some condensation polymers are formed with one type of monomer and others are formed when two different monomers react and connect to each other in an alternating pattern. **Figure 1** generally depicts the reactants and products for a condensation reaction and condensation polymerisation.

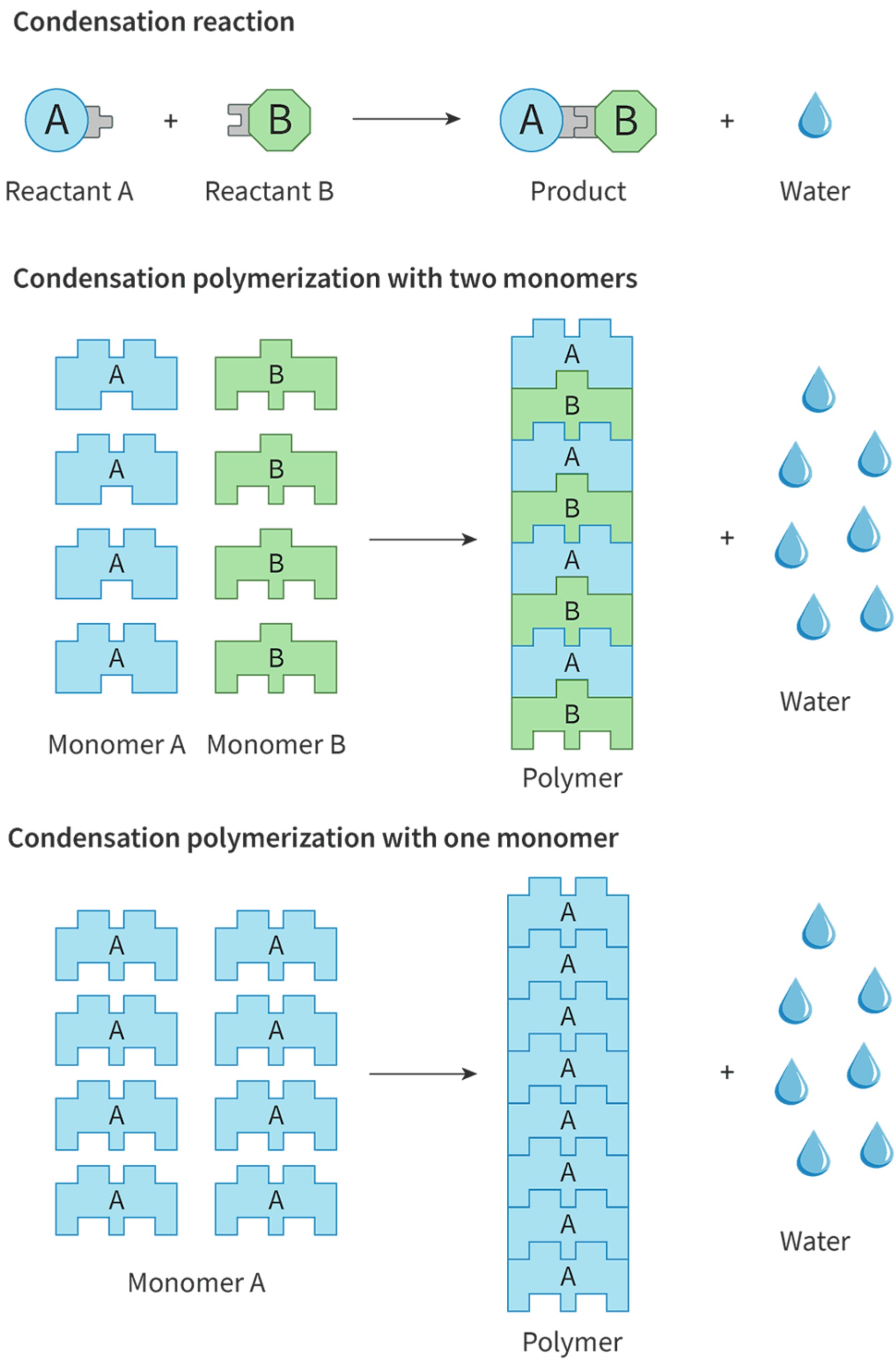


Figure 1. General condensation reaction and condensation polymerisation.

More information for figure 1

The diagram illustrates condensation polymerization showing two different processes. On the top, it starts with two types of monomers, labeled as A and B. The monomers are depicted as puzzle pieces that fit into each other, symbolizing the reaction. Monomer A, represented in blue, and monomer B, in green, react to form a polymer chain shown in the center. Water molecules, depicted as drops, are released as byproducts during this process. The diagram includes arrows showing the process flows from monomers to polymers. Below, another representation shows polymer formation from identical monomers labeled as A forming a uniform chain without alternating types. In both processes, condensation reactions release small molecules (water) as byproducts, which is central to the concept of condensation polymerization.

[Generated by AI]

Polyesters, polyamides and biological macromolecules are all examples of condensation polymers. Condensation polymers are categorised according to the functional groups present on the repeating unit. A functional group is a group of atoms present on a molecule that give characteristic physical and chemical properties to a compound. You will learn about functional groups in greater detail in section S3.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/).

Let’s look at each of these categories of condensation polymers individually.

Polyester

Polyesters are most commonly known for use in fabrics and fibres. Polyesters tend to repel water. Can you think of why that would be useful in clothing? Athletic and outdoor clothing that are prone to getting wet from the environment and sweat are often made from polyester to allow for quick drying. The most common polyester found in clothing and plastics is polyethylene terephthalate (PETE).

A polyester is a polymer formed through condensation polymerisation that contains the ester functional group in its repeating unit. Let’s look at the structure of the polyester PETE and identify its functional group in **Figure 2**.

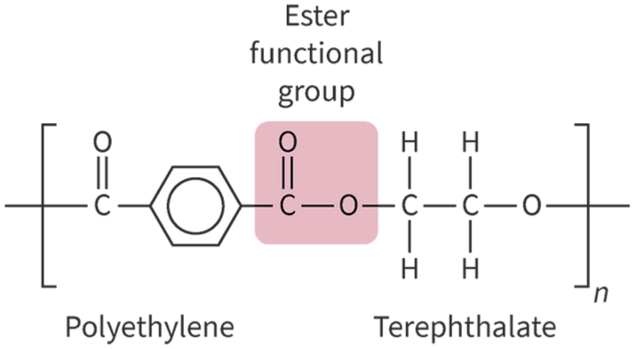


Figure 2. Polyester polymer polyethylene terephthalate.

More information for figure 2

The diagram illustrates the polyester polymer, specifically polyethylene terephthalate (PET), highlighting its ester functional group. The central part of the structure consists of a carbon atom double-bonded to an oxygen atom, and single-bonded to another oxygen atom. This second oxygen atom is further bonded to a second carbon atom, emphasizing the typical ester linkage found in polyester materials.

[Generated by AI]

Notice that the polymer in **Figure 2** has a carbon atom in the middle of the polymer chain that is bonded to two oxygen atoms, one with a double bond and one with a single bond. The oxygen with the single bond is also bonded to another carbon atom. This grouping of atoms makes up the ester functional group. You will learn more about these monomer molecules in section S3.2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/functional-groups-id-43864/). Have you noticed that this polymer is more complex in structure than the addition polymers we looked at in section S2.4.5 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/addition-polymers-id-43846/)? What do you think that means about the monomers that make up polyesters?

The ester functional group can be formed from the combination of functional groups on its reactant molecules. Polyesters are produced from the combination of monomers containing two functional groups on each monomer. **Figure 3** shows the formation of an ester and a polyester from its reactant and monomer molecules.

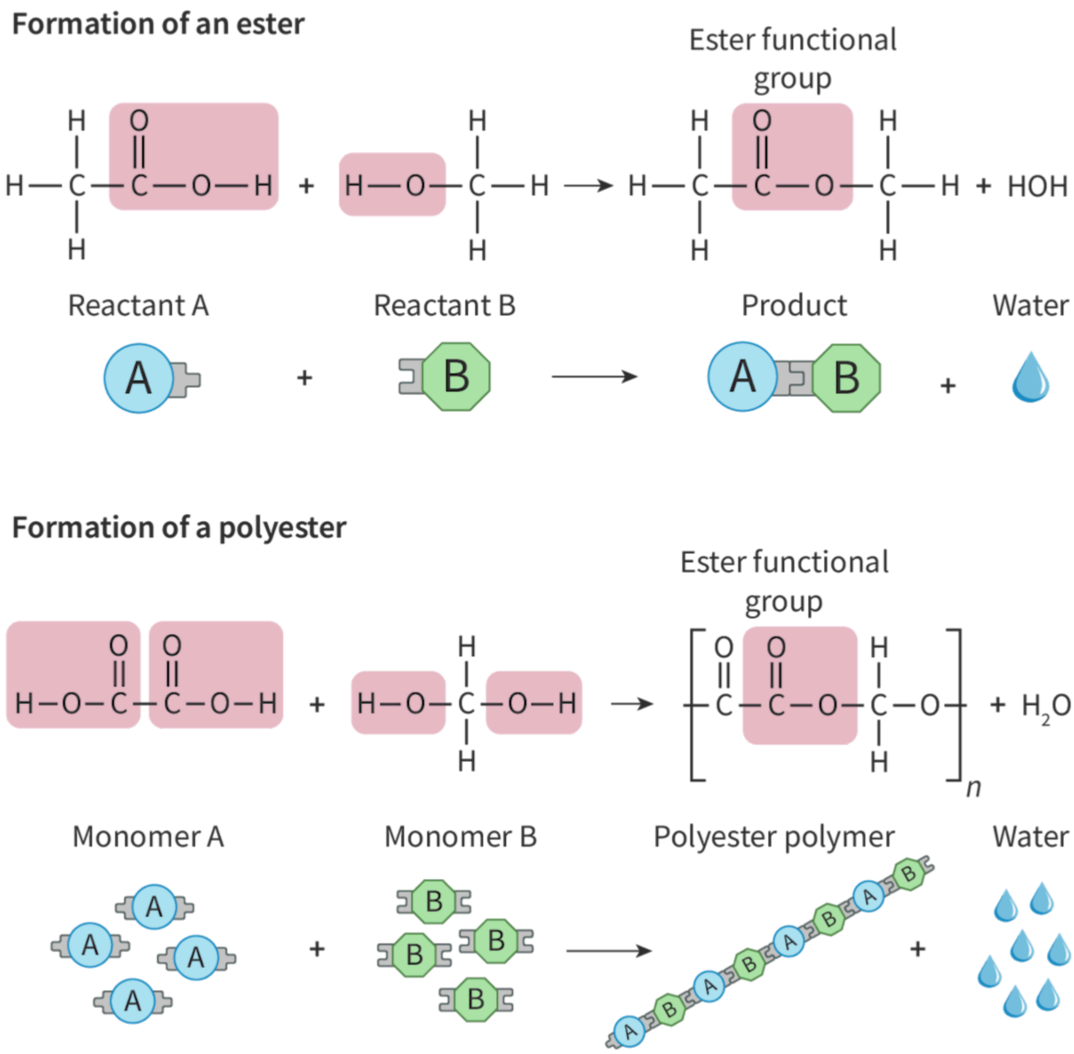


Figure 3. Formation of an ester (top) and a polyester (bottom).

More information for figure 3

The image depicts two chemical reactions: the formation of an ester and the formation of a polyester.

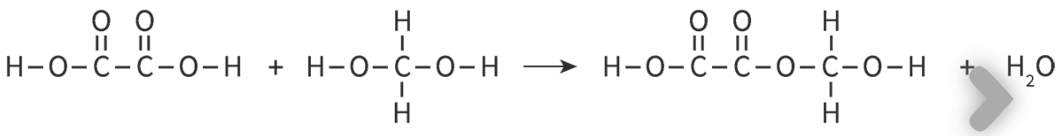
1. Formation of an Ester (top part):
2. The reaction involves two molecules. On the left, "Reactant A" is shown as a molecule with a structural formula, featuring a carbon atom double-bonded to oxygen and single-bonded to an oxygen atom, forming the ester functional group, highlighted in pink. This reactant is combined with "Reactant B", which also displays a structural formula. The two reactants join, forming a "Product" shown as a combination of the two reactants, with an ester functional group at the connection point. A water molecule is also released as a byproduct, indicated by an arrow pointing to an "H2O" symbol.
3. Formation of a Polyester (bottom part):
4. This reaction shows the combination of multiple "Monomer A" molecules and "Monomer B" molecules. Each monomer structure has an ester functional group (highlighted in pink), similar to that in the ester formation. "Monomer A" and "Monomer B" react to form a long chain, the "Polyester polymer", depicted as a repeated sequence of A-B-A-B units. Water

Kog

molecules are represented as byproducts of the reaction, illustrated as multiple water droplets.

[Generated by AI]

Notice in **Figure 3** how water is produced as a byproduct. The release of this small molecule, water, is why reactions of this type are referred to as condensation reactions. **Interactive 1** goes through more specifics about the type of reaction, including the names of some families of compounds you will learn about in more detail in section S3.2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/). Click the arrows to view the slides.



Interactive 1. Formation of a polyester polymer.

More information for interactive 1

A slideshow with six slides displays the Formation of a Polyester Polymer. Slide 1. The chemical reaction between a dicarboxylic acid and a dialcohol. The dicarboxylic acid is represented as a molecule with two carboxylic acid groups $(-COOH)$ on either end.

The structure starts with a hydrogen atom bonded to an oxygen (hydroxyl group), followed by a carbon double bonded to an oxygen (carbonyl group). This carbon is connected to another carbon with the same carboxylic acid structure and its structural formula is

$HOOC{-}COOH$. The acid is labelled as Dicarboxylic acid under it. The dialcohol is depicted as a central carbon atom bonded to two hydroxyl groups $(-OH)$ and two hydrogen atoms. The chemical formula of the alcohol is $HO{-}CH_2{-}OH$. In this case, the central carbon is shown with one hydroxyl on each side. The molecule is labelled as dialcohol under it. An arrow points to the products formed: an ester molecule and water (H_2O). The ester molecule formed by linking the dicarboxylic acid and the dialcohol. The left part of the molecule remains the same as the dicarboxylic acid, but one —OH group

Kog

from the dicarboxylic acid has reacted with the $-OH$ from the dialcohol, forming a new ester bond ($-CO-O-$). The structure preserves the $-OH$ of the diol on the other end. A single water molecule is shown as a byproduct of the condensation reaction.

Slide 2: The $-OH$ from the right side of dicarboxylic acid and the H on the left side of the dialcohol are highlighted. The reaction of the groups leads to the formation of water on the product side, and the water molecule is also highlighted. On-screen text reads: Water is formed from the $-OH$ of the functional group on the carboxylic acid combining with the H of the functional group on the alcohol.

Slide 3. The COOH and OH groups of the ester product are highlighted. On-screen text reads: Notice how the polymer formed still has the functional groups of the alcohol and carboxylic acid. This allows it to react further.

Slide 4. A chemical equation shows the condensation reaction of three organic molecules forming a larger molecule and producing two H_2O molecules. The reactants include three dialcohol, $HO-CH_2-OH$, on the left side, the ester product in the middle and the dicarboxylic acid $HOOC-COOH$ on the right side. The H atom in dialcohol, the $-OH$ group on the left side of ester and H on the right side of ester along with the $-OH$ group on the left side of the dicarboxylic acid are highlighted. The product formed is a polyester after the loss of two water molecules from the reactant molecules. On-screen text reads: Now two H_2O molecules have been produced.

Slide 5. A single ester molecule is shown, consisting of repeating units of carbon, hydrogen, and oxygen atoms. The structure begins and ends with a hydroxyl group ($H-O$) bonded to a carbon atom. The structure is given as, $H-O-CH_2-OCO-COO\text{-}CH_2-OCO\text{-}CO\text{-}O-H$. The overall chain shows multiple ester groups, indicating a polyester structure formed through condensation polymerisation. On-screen text reads: The product formed still has the functional group of an alcohol and carboxylic acid. This allows it to continue to react to form a polymer.

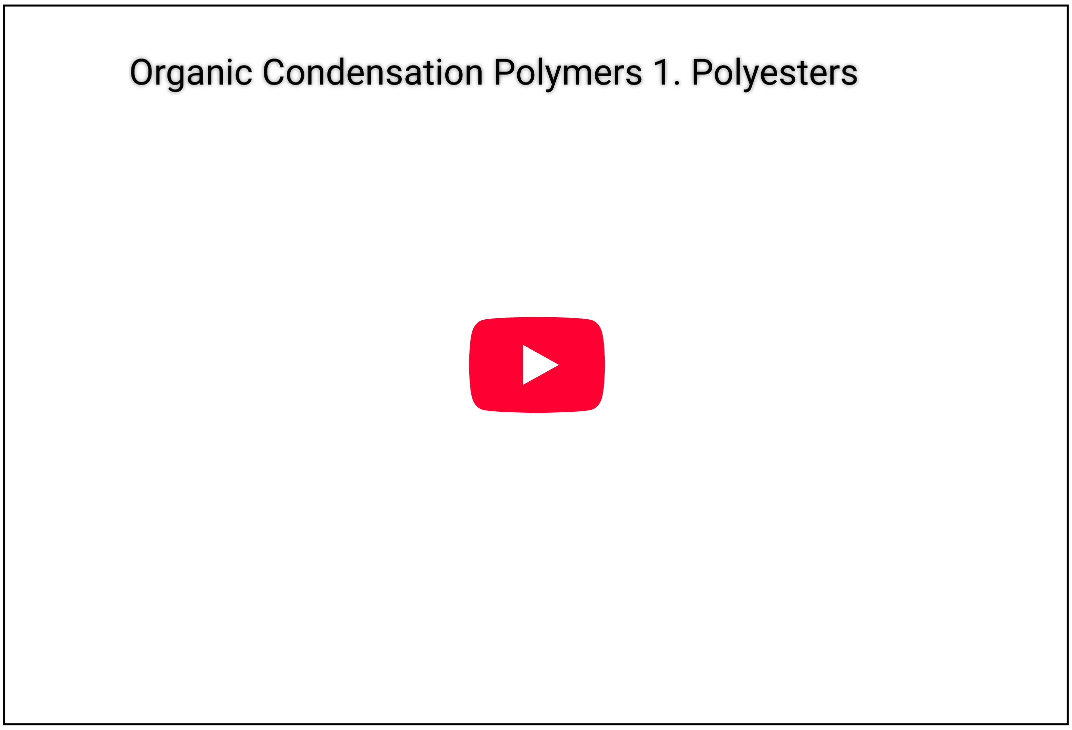
Slide 6. Chemical equation of polyester formation from monomers. On the left side,

$n\ HOOC\text{-}COOH$ under it, the name dicarboxylic acid monomer is written; the next reactant is

$n\ HO\text{-}CH_2\text{-}OH$ under it, the name dialcohol monomer is written. The reaction forms a repeating polymer unit in brackets, labelled as polyester polymer, with linkage $n\ [\text{-}CO\text{-}COO\text{-}CH_2\text{-}O\text{-}]$ and water molecule is represented as $(2n\text{-}1)\ H_2O$ labelled as Water by product.

The interactive helps users understand the steps involved in polyester formation.

The video shows the formation of polyesters in a bit more detail. It is worth revisiting this video once you have studied the family of compounds that make up these monomers section S3.2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/).



Video 1. Formation of polyester polymers.

More information for video 1

1
00:00:00,080 --> 00:00:00,920
[silence]
2
00:00:01,320 --> 00:00:04,880
instructor: Here I take a look
at condensation polymers
3
00:00:05,080 --> 00:00:07,400
or condensation polymerization.
4
00:00:07,960 --> 00:00:10,960
There are two main categories
of condensation polymer,
5
00:00:12,200 --> 00:00:14,760
polyesters and polyamides.
6
00:00:14,840 --> 00:00:17,080
So here I take a look at polyesters.
7
00:00:19,200 --> 00:00:22,280
Polyesters are

made from two monomer units.
8
00:00:22,800 --> 00:00:25,800
Here we see a so-called
dicarboxylic acid.
9
00:00:26,800 --> 00:00:27,920
We have a molecule
10
00:00:28,000 --> 00:00:32,480
that has a carboxylic acid
or carboxyl group at both ends.
11
00:00:33,320 --> 00:00:35,320
The bit that's highlighted in green
12
00:00:35,400 --> 00:00:37,600
between the carboxyl groups
can vary,
13
00:00:38,080 --> 00:00:40,320
so we can have
lots of different possibilities,
14
00:00:40,440 --> 00:00:43,080
but we'd need to have
a carboxyl group at either end
15
00:00:43,280 --> 00:00:46,320
of the dicarboxylic acid, as you'll see.
16
00:00:47,280 --> 00:00:49,560
And the second monomer
is a diol,
17
00:00:49,640 --> 00:00:55,200
a molecule with alcohol or OH
or hydroxyl groups at both end,
18
00:00:55,280 --> 00:00:56,400
so-called diol.
19
00:00:56,720 --> 00:00:59,880
And here again, the bid and blue can vary,
20
00:01:00,000 --> 00:01:02,200
but we must have an OH at both ends.
21
00:01:02,520 --> 00:01:04,240
And this is how they interact.
22

Kog

00:01:04,640 --> 00:01:06,800
The OH of the carboxyl group,
23
00:01:07,360 --> 00:01:10,440
including you notice the bond
between the C and the O
24
00:01:10,720 --> 00:01:14,560
that comes away
from the carboxyl group and the H.
25
00:01:15,040 --> 00:01:20,240
The H only comes away
from the OH of the adjacent diol.
26
00:01:20,800 --> 00:01:22,680
Together they make a molecule of water
27
00:01:23,480 --> 00:01:25,840
and then the two electrons
that represent the bond
28
00:01:25,920 --> 00:01:28,400
that had been between the O
and the H of the hydroxyl group.
29
00:01:28,760 --> 00:01:31,400
Those two electrons are used
to make a bond
30
00:01:31,480 --> 00:01:33,800
with the C of the carboxyl group
31
00:01:34,880 --> 00:01:36,560
that's known as esterification,
32
00:01:36,640 --> 00:01:41,120
the interaction of a carboxyl group
with a hydroxyl group,
33
00:01:41,640 --> 00:01:43,600
a carboxylic acid with an alcohol.
34
00:01:45,960 --> 00:01:48,720
And if we do that
at both ends of these molecules,
35
00:01:48,800 --> 00:01:53,360
in other words,
take an OH as a hydroxide ion off
36

Kog

00:01:54,280 --> 00:01:55,800
both carboxyl groups
37
00:01:56,200 --> 00:01:59,400
and take H as a hydrogen ion off both
38
00:01:59,840 --> 00:02:01,480
hydroxyl groups of the diol.
39
00:02:02,160 --> 00:02:04,200
Then this is what we should have
40
00:02:05,200 --> 00:02:08,400
and let's use those two electrons
of what had been the bond
41
00:02:08,480 --> 00:02:10,280
between the O and the H of the hydroxyl.
42
00:02:10,360 --> 00:02:11,840
To form that linkage,
43
00:02:12,800 --> 00:02:16,600
we're joining up
what had been the CO of the diol
44
00:02:17,120 --> 00:02:20,040
with the C double bond O
of what had been the carboxyl group
45
00:02:20,800 --> 00:02:22,840
and that's known as an ester link.
46
00:02:24,040 --> 00:02:27,880
I've drawn around
the ester linkage C double bond O,
47
00:02:27,960 --> 00:02:31,520
dash O dash C,
which you find in all esters.
48
00:02:32,400 --> 00:02:35,720
And what we do is
we do that on a large scale,
49
00:02:36,920 --> 00:02:40,080
taking a large number
of both types of molecule
50
00:02:40,560 --> 00:02:42,160
and alternating them like this.

Kog

51
00:02:42,320 --> 00:02:43,760
So there I've got
52
00:02:44,080 --> 00:02:45,920
the one from the dicarboxylic acid
53
00:02:46,080 --> 00:02:50,080
and I link up the one
from the diol and keep alternating.
54
00:02:50,160 --> 00:02:53,040
There's another one has come in
from the dicarboxylic acid
55
00:02:53,560 --> 00:02:54,760
and one from the diol.
56
00:02:54,840 --> 00:02:58,720
So you see that we're using both ends
of both molecules,
57
00:02:58,800 --> 00:03:02,320
which is why we need
a dicarboxylic acid and a diol.
58
00:03:05,400 --> 00:03:08,200
Remember the middle bit
there highlighted in green.
59
00:03:08,600 --> 00:03:11,920
That can vary from one
dicarboxylic acid to another.
60
00:03:12,080 --> 00:03:14,600
So it means you can form
lots of different polyesters,
61
00:03:15,000 --> 00:03:18,920
as is the case with the blue bit
highlighted for the diol.
62
00:03:20,520 --> 00:03:22,680
So varying the green,
varying the blue
63
00:03:23,120 --> 00:03:27,040
gives rise to lots
of different polyesters.
64
00:03:27,440 --> 00:03:29,840

Then in the abbreviated
form of this reaction,
65
00:03:29,960 --> 00:03:31,600
as I'm drawing out here,
66
00:03:31,800 --> 00:03:37,520
we take n dicarboxylic acid molecules
and react them with n diol molecules,
67
00:03:38,000 --> 00:03:41,920
where n's a large whole number
or large integer, such as 1 or 2000.
68
00:03:43,040 --> 00:03:45,720
And that means that we'll end up
with n of these
69
00:03:45,800 --> 00:03:48,720
showing how they're joined
through those ester linkages.
70
00:03:50,040 --> 00:03:52,720
Now let's figure out how many
water molecules will be produced.
71
00:03:53,240 --> 00:03:54,600
Think of this analogy.
72
00:03:54,680 --> 00:03:58,280
If you get fence posts,
you have seven spaces between them.
73
00:03:58,880 --> 00:04:02,320
So in this case, hopefully you can see
that if you've got 2n molecules
74
00:04:02,400 --> 00:04:05,680
in total you'll have
2n minus 1 water molecules made.
75
00:04:06,200 --> 00:04:09,520
But notice that I've also taken
effectively another water molecule off
76
00:04:10,240 --> 00:04:13,160
since I've taken an OH off
the left and the H off the right.
77
00:04:13,240 --> 00:04:17,280
So there is actually a total of 2n

Kog

water molecules produced in that scenario.
78
00:04:17,760 --> 00:04:19,800
But if you draw it like this
at the bottom left,
79
00:04:19,880 --> 00:04:21,840
where you've left the OH on the left
80
00:04:21,960 --> 00:04:24,040
and left the H on the extreme right,
81
00:04:24,520 --> 00:04:27,800
then in that scenario,
you do have to indicate
82
00:04:28,200 --> 00:04:31,720
that you get 2n minus 1
water molecules instead.
83
00:04:32,880 --> 00:04:36,080
So that's my look
at the formation of a polyester.
84
00:04:37,000 --> 00:04:38,920
I felt I needed to show both alternatives
85
00:04:39,000 --> 00:04:42,440
because you'll see both
in various textbooks.
86
00:04:44,240 --> 00:04:47,400
One of the accompanying videos takes
a look at a specific polyester
87
00:04:47,480 --> 00:04:48,640
called terylene.
88
00:04:49,280 --> 00:04:50,760
Perhaps you'll consider looking at that.
89
00:04:50,840 --> 00:04:52,280
Thanks for listening!

Polyamide

Polyamides form fibres that are strong allowing it to be used in products such as ropes, seatbelts, camping equipment, carpets and dental floss. Some of the most common polyamides are often referred to as nylon. A polyamide is a polymer formed through condensation polymerisation that contains the amide functional group in its repeating unit. Let's take a look at the structure of nylon-6,6, a common polyamide used in carpets, in **Figure 4** and identify its functional group.

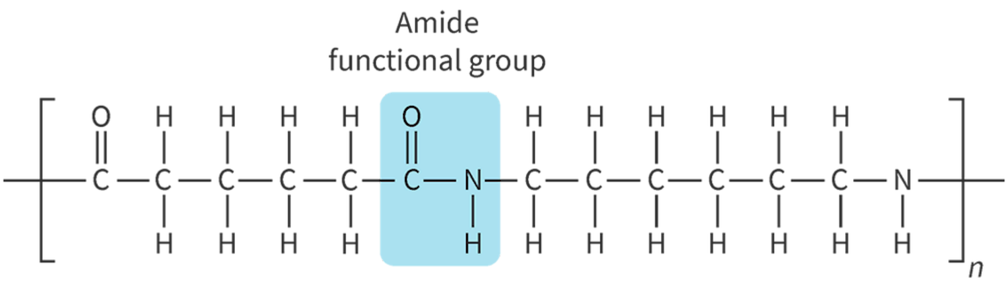


Figure 4. Polyamide polymer nylon-6,6.

More information for figure 4

The image displays the chemical structure of the nylon-6,6 polymer. The structure is represented in a linear format showing repeating units where the amide functional group is highlighted. This functional group consists of a carbon atom doubly bonded to an oxygen atom and singly bonded to a nitrogen atom, which in turn is bonded to a hydrogen atom. The carbon atom is also part of a chain link with other carbon atoms, forming the backbone of the polymer. The repeating units emphasize the polyamide characteristic of nylon-6,6, displaying the connectivity that allows it to form long, strong fibers useful in various products such as ropes and carpets.

[Generated by AI]

Notice that the polymer in **Figure 4** has a carbon atom in the middle of the polymer chain that is doubly bonded to one oxygen atom and singly bonded to a nitrogen atom. This grouping of atoms makes up the amide functional group. Can you think of any reason why the polymer in **Figure 4** is known as nylon-6,6?

Notice how each side of the nitrogen functional group is bonded to a chain containing 6 carbon atoms? This is where the naming of different types of nylon come from.

Polyamides are produced in a very similar manner to polyesters requiring monomers that contain two functional groups. The functional groups on one of the monomers differs from those that make up polyesters. **Figure 5** shows the formation of an amide and a polyamide from its reactant and monomer molecules.

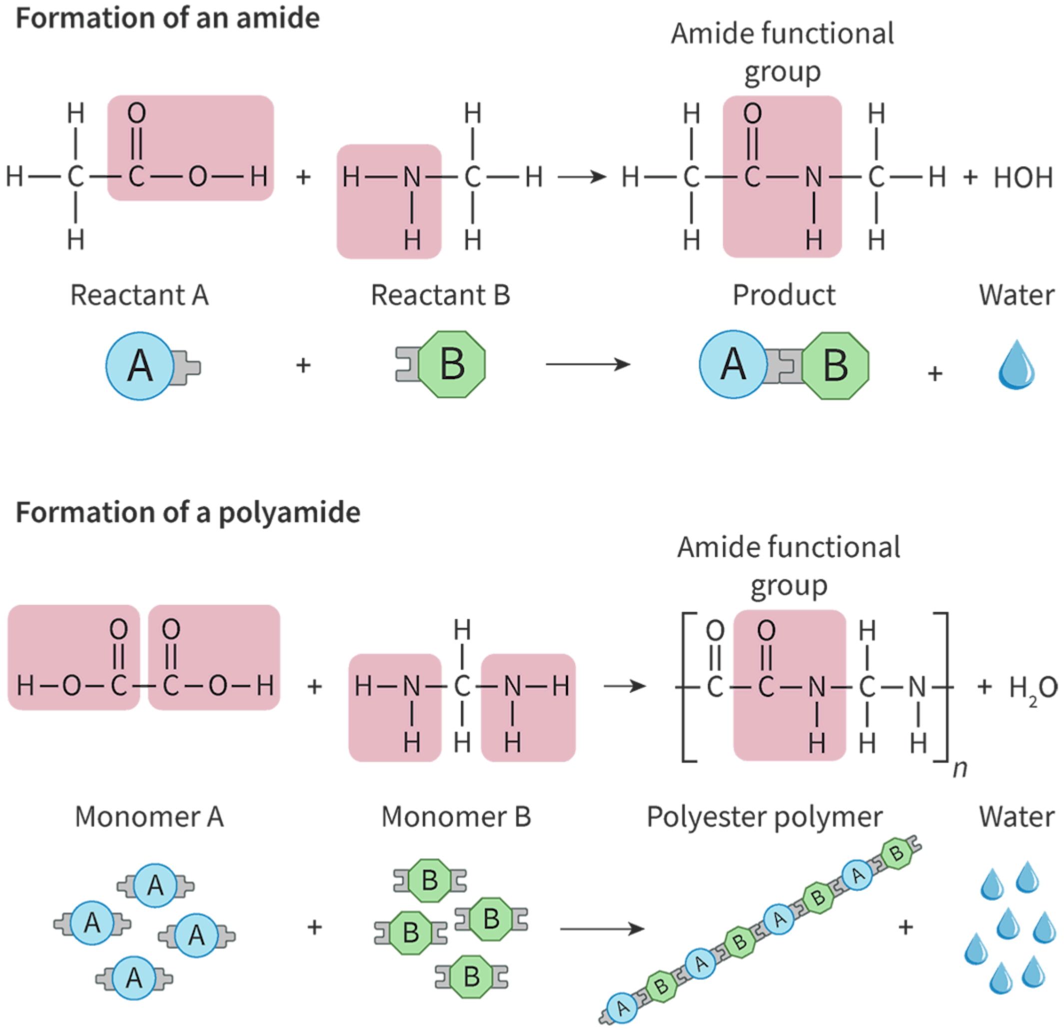


Figure 5. Formation of an amide and a polyamide.

More information for figure 5

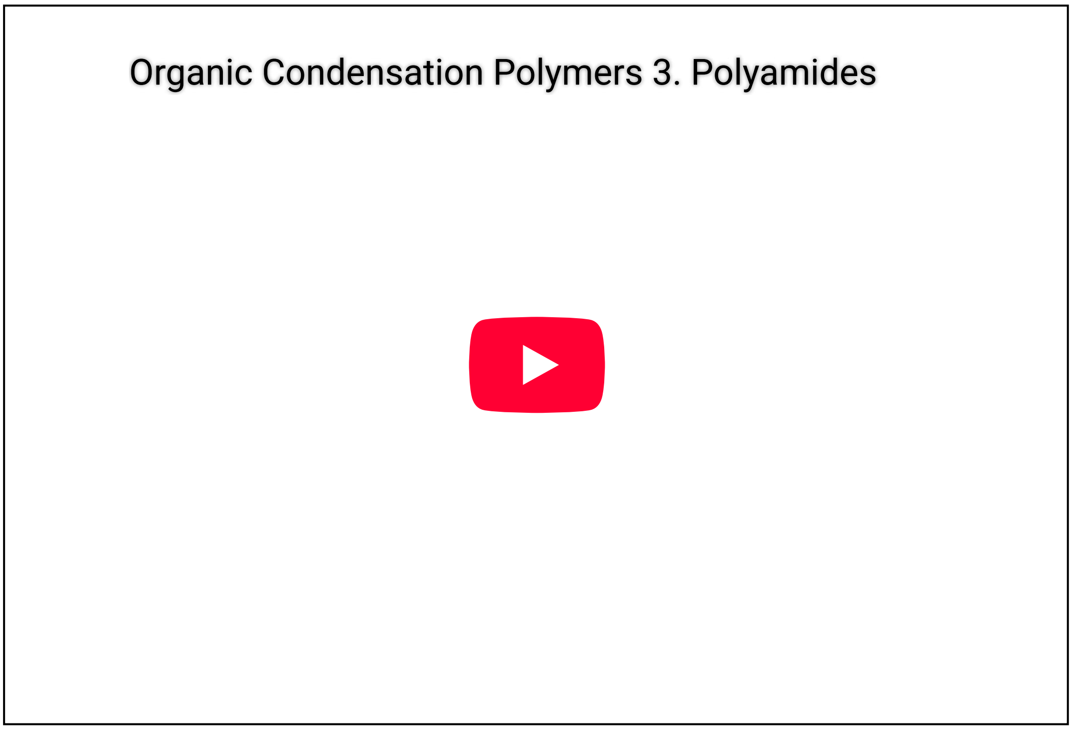
The diagram illustrates the chemical process of forming an amide and a polyamide from monomers. It is organized into three main sections, showing both molecular structures and the reactions involved.

1. **Amide Formation:**
2. On the top left, a chemical structure is depicted showing an amide bond formation between a carboxylic acid group (O=C-OH) and an amine group (NH2). The connection releases water (depicted by a water droplet icon) as a byproduct.
3. **Polyamide Formation:**
4. On the right, two different monomers labeled as 'A' and 'B' each possess distinct functional groups. Monomer 'A' has a blue circle with an arrow indicating it connects with monomer 'B', which is shown in a green pentagon.
5. The structure displayed in the mid-section depicts the repeated sequence of amide bonds forming a chain, denoting polyamide structure. Arrows indicate the continuous sequence of reactions, creating repeating units (A-B-A-B). Every connection releases a water molecule during the process.
6. **Polyamide Chain Representation:**

7. The bottom section further elaborates the polyamide formation by showing elongated chains of monomers. Each chain consists of alternating 'A' and 'B' units, emphasizing the polymer's repeating structure.

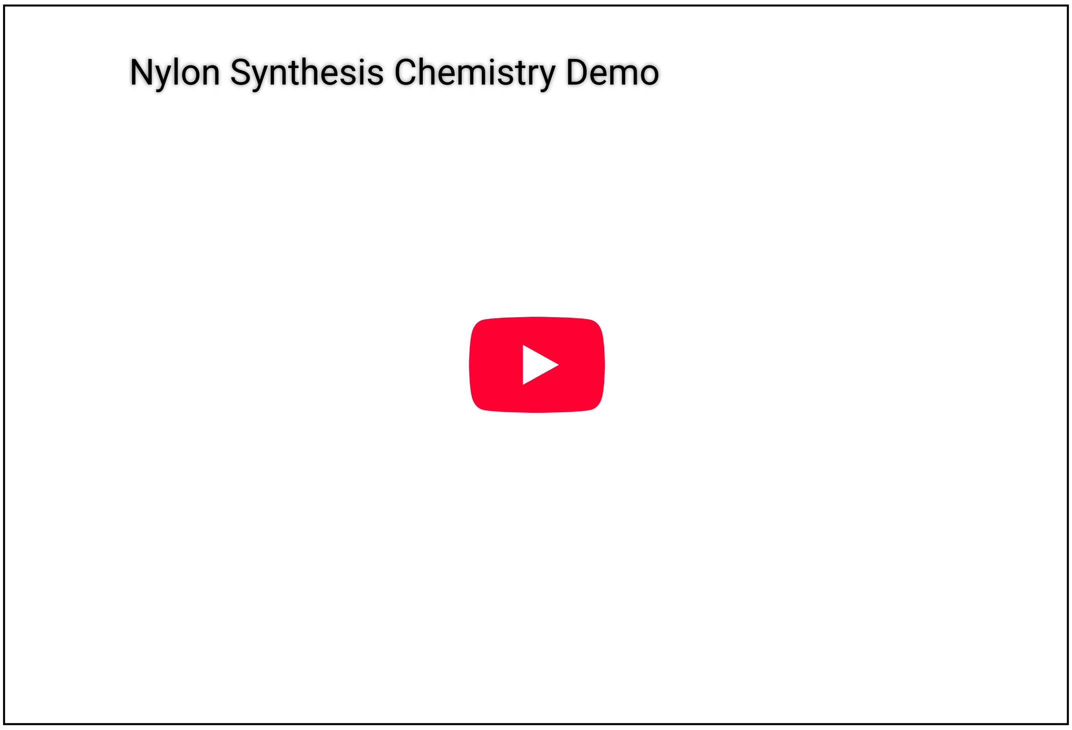
[Generated by AI]

The video shows the formation of polyamides in a bit more detail. It is worth revisiting this video once you have studied the family of compounds that make up these monomers in section S3.2.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/homologous-series-id-43865/).



Video 2. Formation of polyamide polymers.

The video is an example of producing a polyamide polymer, nylon-6,10 in a laboratory setting.



Video 3. Production of polyamide nylon-6,10.

Theory of Knowledge

The widespread use of synthetic fibres in the clothing industry began in the 1940s as a more durable and cost-effective alternative to natural fibres. Greater durability also means greater resistance to breaking down, resulting in millions of tonnes of global textile waste each year as clothing styles go in and out of fashion. To what extent is the creator and the user of this technology responsible for the impact on the environment?

Biological macromolecules

All biological macromolecules are formed through condensation reactions. Some examples of biological macromolecules that are also polymers are:

- polypeptides (proteins)
- polysaccharides (sugars)
- polynucleotides (nucleic acids such as DNA).

The structure of these polymers and their monomers are shown in **Figure 6**.

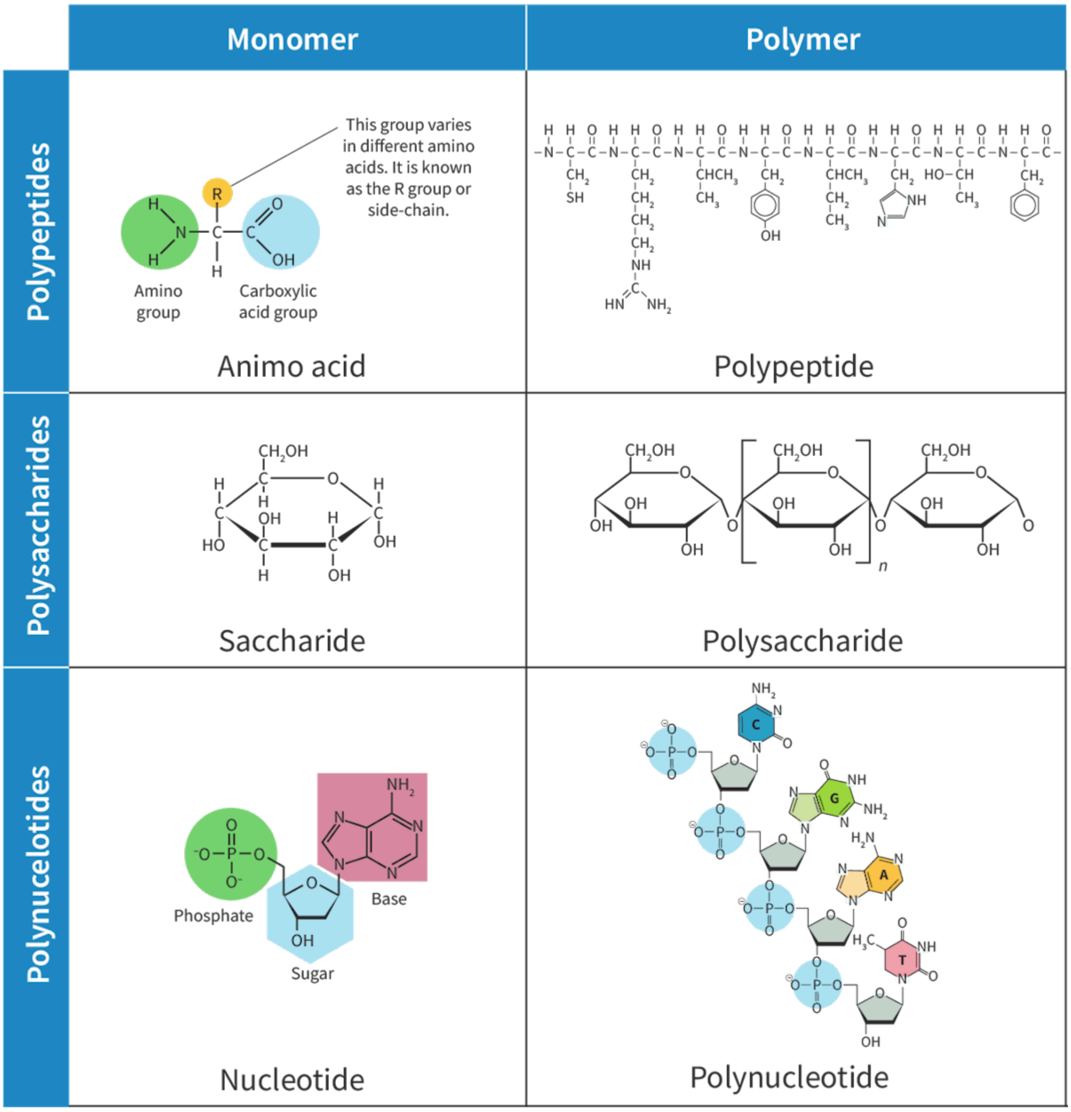


Figure 6. Monomer and polymer structures for some biological macromolecules.

More information for figure 6

The image is a diagram showcasing the structures of monomers and polymers for three types of biological macromolecules: polypeptides, polysaccharides, and polynucleotides. The diagram is divided into three rows, each representing one type of macromolecule.

1. Polypeptides:
2. Monomer: The left section shows an amino acid structure featuring an amino group and carboxylic acid group with a variable R group (side chain).
3. Polymer: The right section displays a polypeptide chain with repeated units of amino acids connected by peptide bonds.
4. Polysaccharides:
5. Monomer: The left section shows a saccharide with a basic carbon ring structure.
6. Polymer: The right section presents a polysaccharide chain with multiple saccharides linked together.

7. Polynucleotides:
8. Monomer: The left section features a nucleotide with a phosphate group, sugar, and base, highlighted in different colors for each part.
9. Polymer: The right section illustrates a polynucleotide chain with repeated sequences of nucleotides forming a DNA structure equivalent.

Each section is clearly labeled, indicating the monomer and polymer of each macromolecule type.

[Generated by AI]

Polypeptides

Polypeptides are polymers that make up proteins from monomers called amino acids. The general structure for an amino acid monomer is shown in **Figure 6**. Looking at the general structure of amino acids, what do you notice? Does the structure contain functional groups that we have already seen so far? Notice how the amino acid has two different functional groups. This means that we do not need two different types of monomers to create a polymer as we saw earlier, we can use just one monomer to create a polymer. Looking at the structure of the amino acid monomer, would you expect amino acids to form polyesters or polyamides? Let's look at the formation of a polypeptide in **Figure 7**.

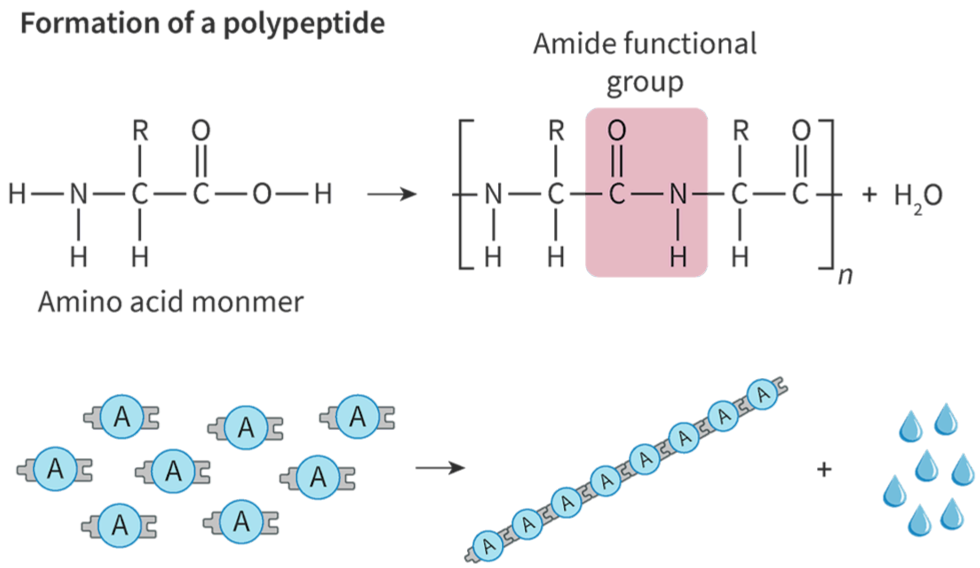


Figure 7. Condensation reaction to form a polypeptide.

More information for figure 7

The image is a diagram depicting a condensation reaction forming a polypeptide. On the left side of the diagram, individual amino acid molecules are illustrated, each represented by the letter 'A' inside blue circles, and attached to a rectangular shape representing the amino acid structure. The amino acids are shown aligned horizontally, indicating the start of a forming chain.

Arrows pointing from each amino acid toward a central bond formation area visually indicate the process of linking these amino acids together. In the center, a pink area highlights the formation of an amide bond (depicted by the chemical structure O=C-N-H), signifying the creation of a polypeptide bond between the amino acids.

To the right, the newly formed polypeptide chain is shown, represented by multiple 'A' circles connected in a linear sequence to emphasize the continuous structure of the polymer. Above the polypeptide, water molecules are illustrated as byproducts of the reaction, shown as small blue droplet icons.

This whole process demonstrates how a polypeptide forms as a polymer from amino acids via condensation reaction, with water being released as a byproduct, illustrated by the droplets and chemical bond formation.

[Generated by AI]

You can see in **Figure 7** that when amino acids combine to form a polypeptide polymer, the amide functional group is formed, making the polypeptide polymer an example of a polyamide. Notice how water is also produced as a byproduct during this condensation reaction. Proteins can contain polypeptides formed from anywhere from 11 to 35 000 amino acids.

Polysaccharides

Polysaccharides are the polymers that make up sugars, such as starch and cellulose. Monomers called saccharides combine through a condensation reaction to form a polysaccharide and water. The structures of the polysaccharide starch and its saccharide glucose monomer are shown in **Figure 6**.

Polynucleotides

Polynucleotides are genetic material, such as your DNA, that form by a condensation reaction of monomers called nucleotides. The structures of a polynucleotide and its nucleotide monomer are shown in **Figure 6**. Nucleotides are made up of a base, a sugar and a phosphate component. The reacting functional groups are on the sugar and phosphate components. The genetic code is contained in the sequences of bases. DNA consists of two strands of polynucleotides twisted together and attracted by hydrogen bonds with the bases on the inside to protect the code. This is illustrated in **Figure 8**.

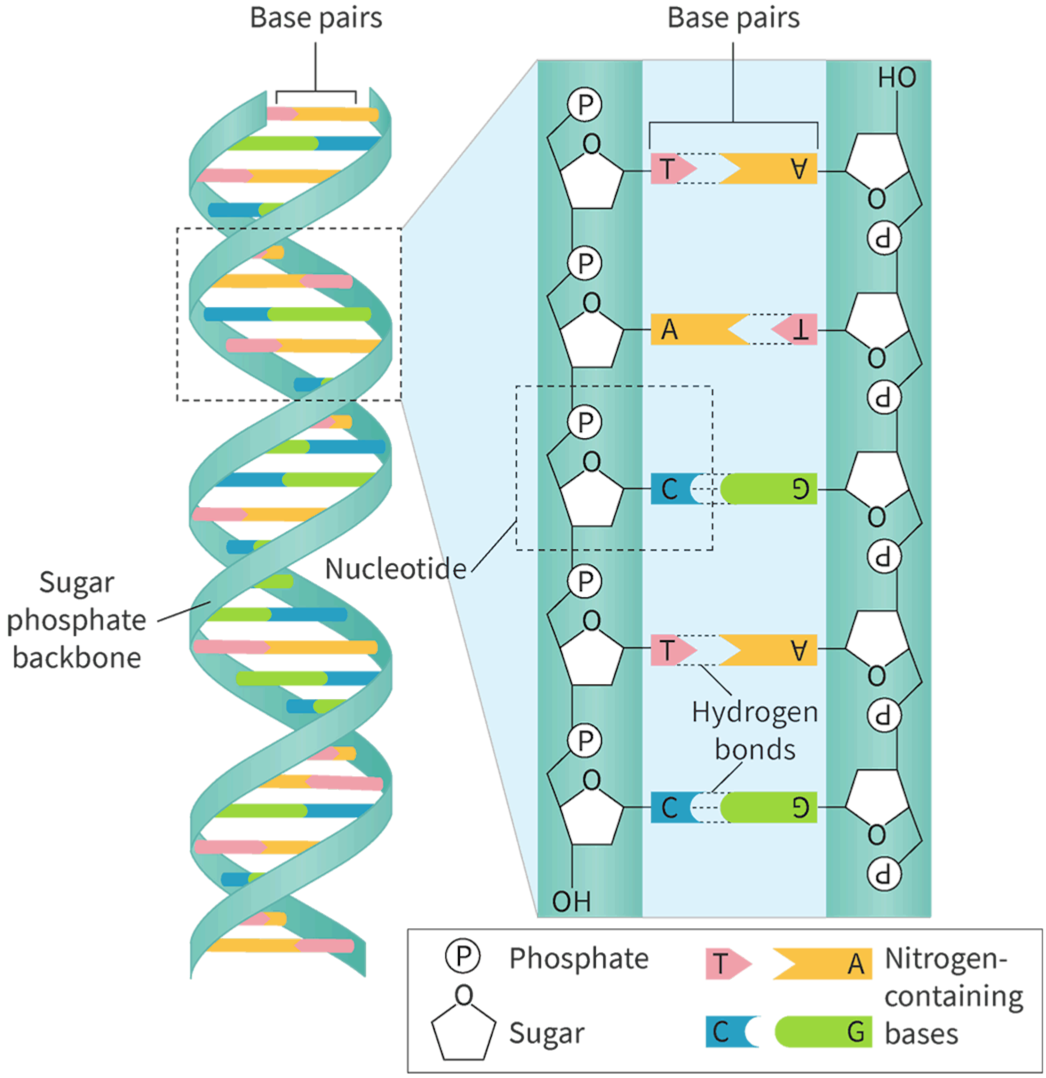


Figure 8. The structure of DNA.

More information for figure 8

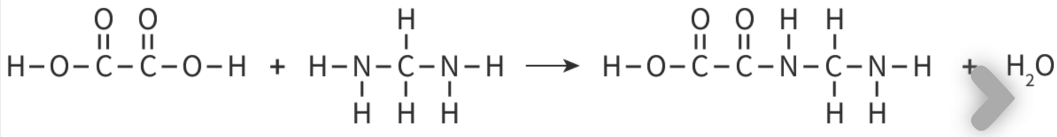
The image shows a detailed diagram of DNA structure, featuring both the double helix and a closer view of nucleotide bonding. On the left side of the image, the double helix structure is depicted, illustrating how two strands twist around each other. Each strand appears as a series of interconnected segments with bases displayed inside.

On the right side, a magnified view shows the bonding between nucleotides, which consist of a sugar (represented by a pentagon), a phosphate group (marked "P"), and a base (one of A, T, C, or G). Bases are paired across the two strands: Adenine (A) pairs with Thymine (T), and Cytosine (C) pairs with Guanine (G) through hydrogen bonds. These pairs are illustrated with complementary shapes and colors.

A key explains the components during base pairing, and the structural arrangement within the double helix is emphasized to show how the sequence of nucleotides forms the genetic code, protected by the twisting strand structure.

[Generated by AI]

All biological macromolecules can be broken down by a process called hydrolysis. Hydrolysis is the process of breaking a molecule into two parts using a water molecule. The slides in **Interactive 2** show the processes for condensation and hydrolysis for polypeptides, polysaccharides and polynucleotides.



Interactive 2. Processes for condensation and hydrolysis for polypeptides, polysaccharides and polynucleotides.

More information for interactive 2

The slideshow has six slides for the Processes for Condensation and Hydrolysis of Various Polymers.
Slide 1: A chemical equation showing the reaction between a dicarboxylic acid and a diamine. The dicarboxylic acid is represented as a molecule with two carboxylic acid groups $(-COOH)$ on either end. The structure starts with a hydrogen atom bonded to an oxygen (hydroxyl group), followed by a carbon double bonded to an oxygen (carbonyl group). This carbon is connected to another carbon with the same carboxylic acid structure, and its structural formula is $HOOC{-}COOH$. The acid is labelled as Dicarboxylic acid under it. The diamine is depicted as a central carbon atom bonded to two amine groups $(-NH2)$ and two hydrogen atoms. The chemical formula of the amine is $H2N{-}CH2{-}NH2$. In this case, the central carbon is shown with one amine on each side. The molecule is labelled as diamine under it. The product is given as $NH2 - CO - COO - CH2 - NH2$ along with a water molecule. Each time an amine group reacts with a carboxylic acid group, an amide bond $(-CONH-)$ is formed, and a molecule of water $(H2O)$ is released as a by-product.
Slide 2: The $-OH$ group on dicarboxylic acid and H on amine on the reactant side and water molecule on the product side are highlighted. The reaction is the same as the previous slide, and the formation of water molecule is highlighted. On-screen text reads: Water is formed from the $-OH$ of the functional group on the carboxylic acid combining with the H of the functional group on the amine.

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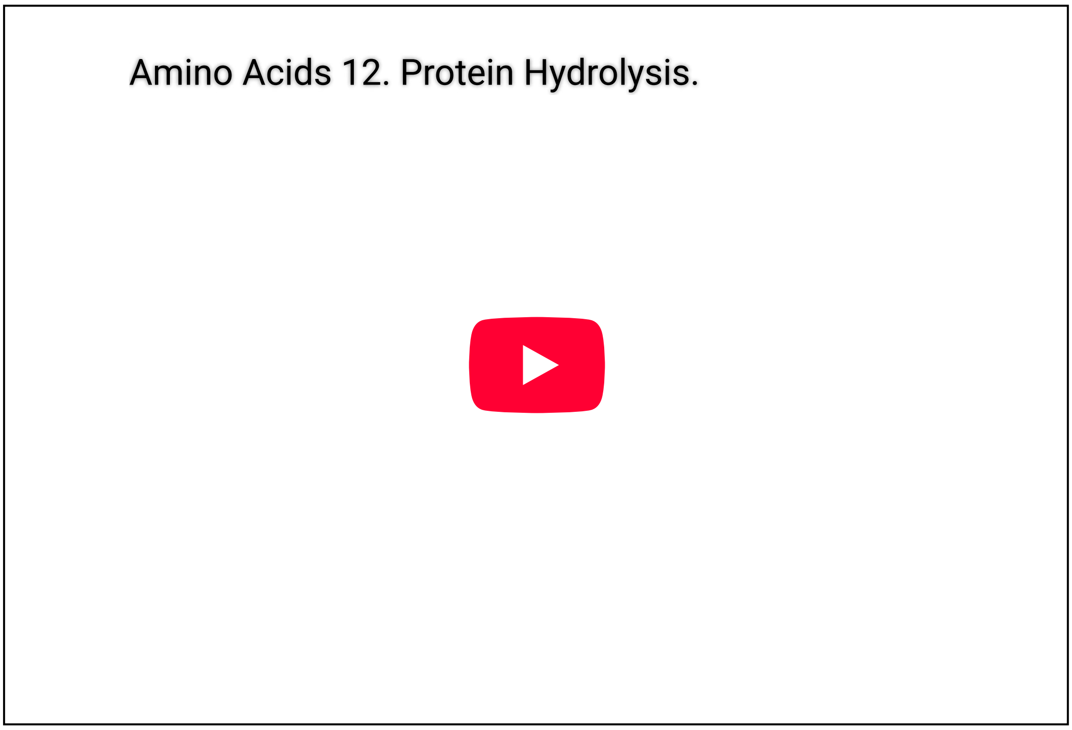
Slide 3: A chemical equation showing polymer formation from dicarboxylic acid and diamine with reactive functional groups is highlighted in the product. The structural formula of the product is $H_2N - CO - CO - NH - CH_2 - NH_2$. The carboxylic acid group and the amine group on the product are highlighted in pink to indicate the presence of remaining reactive sites that enable further polymerisation. On-screen text reads: Notice how the polymer formed still has the functional group of the amine and carboxylic acid. This allows it to react further.
Slide 4: The reaction shown involves the formation of a polyamide through a process called condensation polymerisation. In this reaction, monomers containing amine $(-NH_2)$ groups and carboxylic acid $(-COOH)$ groups react together with an amide. As shown in the first diagram, when the three molecules combine, two water molecules are produced, highlighted by the removal of H and $-OH$ groups. The resulting product is a chain of repeating units joined by amide bonds, with functional amine and carboxylic acid groups still present at the ends. This allows the molecule to continue reacting with additional monomers to form a longer polymer chain. On-screen text reads: In total now 3 H_2O molecules have been produced.
Slide 5: It highlights these reactive end groups, amine and carboxylic acid, demonstrating the potential for further polymer growth. On-screen text reads: The product formed still has the functional groups of an amine and carboxylic acid. This allows it to continue to react to form a polymer.
Slide 6. A dicarboxylic acid monomer reacts with a diamine monomer to form a polyamide polymer and water. The general reaction can be summarised as *n* dicarboxylic acid molecules reacting with n diamine molecules to form a polyamide and *(2n minus 1)* water molecules.

This video shows more detail about hydrolysis of polypeptides.



Video 4. Hydrolysis of polypeptides.

More information for video 4

1
00:00:00,880 --> 00:00:03,280
narrator:
Here I take a look at protein hydrolysis.
2
00:00:03,800 --> 00:00:05,480
By definition, hydrolysis means
3
00:00:05,560 --> 00:00:08,280
to split a molecule apart
using the elements of water.
4
00:00:09,080 --> 00:00:13,040
Typically, an alkali or an aqueous acid
is used to bring this about.
5
00:00:14,080 --> 00:00:16,280
Now as hydrolysis reactions go,
6
00:00:16,360 --> 00:00:19,240
the hydrolysis
of proteins are relatively difficult.
7
00:00:19,880 --> 00:00:22,560
Remember, proteins have at least a primary
8
00:00:22,720 --> 00:00:28,120
and secondary structure and often tertiary
and indeed quaternary structure.
9
00:00:28,920 --> 00:00:32,240
The first of two methods
to bring about protein hydrolysis
10
00:00:32,320 --> 00:00:35,720
would be to reflux the protein
with six molar hydrochloric acid
11
00:00:35,800 --> 00:00:37,000
for about 24 hours.
12
00:00:37,840 --> 00:00:42,680
And the second method
would be to use protease enzymes
13
00:00:43,160 --> 00:00:44,840
over a period of several hours.
14

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00:00:45,000 --> 00:00:47,880
At a line body temperature,
37 degrees centigrade.
15
00:00:48,480 --> 00:00:51,200
And of course, the reason
that it takes such a long time
16
00:00:51,280 --> 00:00:54,440
and a lot of effort to bring
about the hydrolysis of a protein
17
00:00:55,120 --> 00:00:57,080
is because of its complex structure.
18
00:00:57,280 --> 00:00:59,880
You're going to have to break
down any quaternary structure,
19
00:01:00,320 --> 00:01:04,960
any tertiary structure,
any secondary structure
20
00:01:05,360 --> 00:01:08,240
before you get to this point
where you've got your polypeptide chain.
21
00:01:08,320 --> 00:01:11,040
So you're going to have
to break all those bonds
22
00:01:11,120 --> 00:01:15,200
that cause the twisting and folding,
namely Van der Waals' forces,
23
00:01:15,560 --> 00:01:20,000
electrostatic attractions,
hydrogen bonds, disulfide bonds
24
00:01:20,080 --> 00:01:21,640
before you get to this level.
25
00:01:22,320 --> 00:01:23,520
But what I'm going to show then
26
00:01:23,600 --> 00:01:25,320
is what happens
once you've got to this level.
27
00:01:25,400 --> 00:01:29,240
This is the hydrolysis

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of the peptide links
28
00:01:29,720 --> 00:01:32,200
I'm showing there
with the scissors that the bond
29
00:01:32,280 --> 00:01:34,640
between the CO and the NH bricks.
30
00:01:34,920 --> 00:01:38,080
We put an OH to the left
giving us back the carboxyl group,
31
00:01:38,560 --> 00:01:41,400
and we put a H to the right,
giving us back our NH2.
32
00:01:41,800 --> 00:01:44,960
So there's my first amino acid
molecule produced
33
00:01:45,080 --> 00:01:47,480
and this process
is repeated over and over.
34
00:01:47,720 --> 00:01:50,120
So let's bring in the scissors
35
00:01:50,360 --> 00:01:54,160
and bring about the hydrolysis
of the next peptide link.
36
00:01:54,680 --> 00:01:56,400
So again, this bond between
37
00:01:56,760 --> 00:01:59,120
the C double bond O
and the NH is going to break.
38
00:01:59,760 --> 00:02:02,520
We'll put the OH together
with the C double bond O
39
00:02:02,600 --> 00:02:04,200
to give back the carboxyl group.
40
00:02:04,720 --> 00:02:06,640
And we've got another amino acid there.
41
00:02:06,720 --> 00:02:10,840
And we'll put a H along with the NH

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to give us back our NH2.
42
00:02:11,760 --> 00:02:13,760
Now we've got two molecules of amino acid.
43
00:02:14,440 --> 00:02:17,840
Let's bring the scissors
in and snip another peptide link.
44
00:02:18,480 --> 00:02:22,080
The peptide link is made up
of the CO and the NH by the way.
45
00:02:22,960 --> 00:02:25,720
But in hydrolysis,
I'm showing this the bond
46
00:02:25,800 --> 00:02:27,320
between the C and the N break.
47
00:02:27,960 --> 00:02:31,160
And we've got another,
a third molecule of amino acids.
48
00:02:32,320 --> 00:02:36,640
So we've already broken
the quaternary structure,
49
00:02:36,720 --> 00:02:39,960
if it exists, we've broken
the tertiary structure if it exists.
50
00:02:40,040 --> 00:02:42,560
Remember, fibrous proteins
don't really have a tertiary
51
00:02:42,640 --> 00:02:44,240
or indeed quaternary structure.
52
00:02:44,600 --> 00:02:47,560
Globular proteins
do have a tertiary structure
53
00:02:47,760 --> 00:02:50,080
and often a quaternary structure as well.
54
00:02:51,560 --> 00:02:56,320
And what's holding these twisted
and folded structures together
55
00:02:56,680 --> 00:02:58,760

Kog

are generally four kinds of bonds,
56
00:02:59,680 --> 00:03:02,840
namely Van der Waals'
forces between, for example,
57
00:03:03,040 --> 00:03:06,120
alkyl R groups, electrostatic attractions,
58
00:03:06,200 --> 00:03:09,040
for example,
between lysine and aspartic acid,
59
00:03:09,480 --> 00:03:13,160
hydrogen bonds between,
for example, serine and aspartic acid
60
00:03:13,560 --> 00:03:16,880
and disulfide bonds
between cystine molecules.
61
00:03:17,920 --> 00:03:20,880
And we've nearly finished snipping these
62
00:03:21,240 --> 00:03:23,440
peptide links bringing about hydrolysis.
63
00:03:23,520 --> 00:03:24,800
There's our last one,
64
00:03:25,760 --> 00:03:28,840
and again,
we put an OH on the C double bond O.
65
00:03:28,920 --> 00:03:30,680
We put a H on the NH,
66
00:03:31,720 --> 00:03:36,440
and we brought about hydrolysis
creating 1, 2, 3, 4, 5, 6,
67
00:03:36,800 --> 00:03:39,320
now amino acid molecules.
68
00:03:40,280 --> 00:03:43,000
Finally, let's see how we could summarize
69
00:03:43,080 --> 00:03:45,880
this hydrolysis of the polypeptide chain.
70
00:03:46,840 --> 00:03:48,400

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So if we take a protein
71
00:03:48,680 --> 00:03:52,960
and be represented
as n amino acids joined together,
72
00:03:53,200 --> 00:03:55,920
let's count how many
water molecules we'd need.
73
00:03:56,360 --> 00:03:59,040
Well, if you had
three amino acids joined together,
74
00:03:59,760 --> 00:04:03,320
you would require two water molecules
because you would've two peptide links.
75
00:04:03,520 --> 00:04:06,120
So the number of water molecules
is one less
76
00:04:06,200 --> 00:04:08,560
than the number
of amino acids in the chain.
77
00:04:09,320 --> 00:04:15,040
But what you can do is instead of having
awkwardly used n minus 1 water molecules,
78
00:04:15,280 --> 00:04:17,640
you can have n water molecules used,
79
00:04:18,000 --> 00:04:20,040
if you take off
one additional water molecule.
80
00:04:20,120 --> 00:04:22,280
And I've done that by leaving the H off
81
00:04:22,360 --> 00:04:24,360
outside the bracket to the left
82
00:04:24,440 --> 00:04:27,160
and the OH off outside
the bracket to the right.
83
00:04:27,600 --> 00:04:30,040
So that's been my look
at protein hydrolysis.
84

Kog

00:04:30,320 --> 00:04:32,800
Do please consider looking
at others in the series,
85
00:04:33,000 --> 00:04:34,280
amino acids and proteins.

In this next activity you will use a folded piece of paper to model the formation of condensation polymers from their monomers. You will also design your own paper model for a condensation polymer.

Activity

- **IB learner profile attribute:** Thinkers
- **Approaches to learning:** Thinking skills — Combining different ideas in order to create new understandings
- **Time required to complete activity:** 20—30 minutes
- **Activity type:** Pairs activity

Print out the following page . If you do not have a printer request that your teacher print it out for you.

Page
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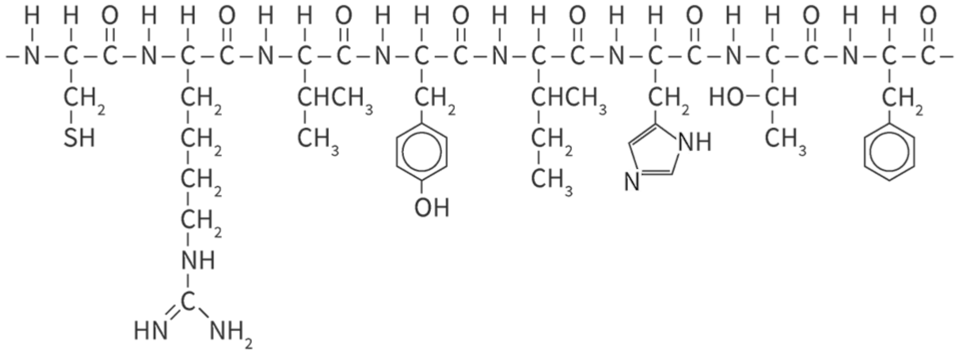
Cut out each polymer into its own strip. Each partner gets their own strip. Perform the following:

- Identify all of the functional groups in the monomer. Share this identification with your partner.
- Fold along the dotted lines to 'condense' the monomers into a polymer.
- Classify the polymer as either a polyester or polyamide. Share your classification with your partner.
- Deduce how many water molecules were condensed from the reaction of this number of monomers.
- Individually, make your own strip out of one of the polymers discussed in this section. You should select a different polymer than your partner.
- Together make a strip to represent the polypeptide shown below.



More information

This is a flowchart diagram depicting a sequence of steps. It starts from a leftmost node labeled 'A', which connects to a middle node labeled 'B'. The sequence continues from 'B' to a rightmost node labeled 'C'. Arrows indicate the direction of the flow from one node to the next. Each node represents a step in the process, with 'A' as the initial step, followed by 'B', and concluding with 'C'. The diagram does not include decision points or alternative paths but shows a straightforward progression of steps. The arrangement is linear, illustrating a simple, one-directional flow.

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